

Chapter 28 Check

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28 Supersymmetric Versions of the Standard Model

Physical phenomena at energies accessible in today’s accelerator laboratories are accurately described by the standard model, the renormalizable theory of quarks, leptons, and gauge bosons, governed by the gauge group $SU(3) \cdot SU(2) \cdot U(1)$, described in Sections 18.7 and 21.3. The standard model is today usually [1] understood as a low-energy approximation to some as-yet-unknown fundamental theory in which gravitation appears unified with the strong and electroweak forces at an energy somewhere in the range of 10^{16} to 10^{18} GeV. This raises the *hierarchy problem*: what accounts for the enormous ratio of this fundamental energy scale and the energy scale ≈ 300 GeV that characterizes the standard model?

The strongest theoretical motivation for supersymmetry is that it offers a hope of solving the hierarchy problem. Quarks, leptons, and gauge bosons are required by the $SU(3) \cdot SU(2) \cdot U(1)$ gauge symmetry to appear with zero masses in the Lagrangian of the standard model, so that the physical masses of these particles are proportional to the electroweak breaking scale, which in turn is proportional to the mass of the scalar fields responsible for the electroweak symmetry breakdown. The crux of the hierarchy problem [1a] is that the scalar fields, unlike the fermion and gauge boson fields, are not protected from acquiring large bare masses by any symmetry of the standard model, so it is difficult to see why their masses, and hence all other masses, are not in the neighborhood of 10^{16} to 10^{18} GeV.

It has been hoped that this problem could be solved by embedding the standard model in a supersymmetric theory. If the scalar fields appear in supermultiplets along with fermions in a chiral representation of some gauge group, then supersymmetry would require vanishing bare masses for the scalars as well as the fermions. All the masses of the standard model would then be tied to the energy scale at which supersymmetry is broken. The hope of a solution of the hierarchy problem along these lines has been the single strongest motivation for trying to incorporate supersymmetry in a realistic theory.

Unfortunately, none of the new particles required by supersymmetric theories have been detected, and no entirely satisfactory supersymmetric version of the standard model has emerged so far. This chapter will describe the attempts that have been made in this direction.

28.1 Superfields, Anomalies, and Conservation Laws

In this section we shall attempt to decide at least tentatively what ingredients should appear in a supersymmetric version of the standard model.

None of the quark and lepton fields of the standard model belong to the adjoint representation of the $SU(3) \cdot SU(2) \cdot U(1)$ gauge group, so they cannot be the superpartners of known gauge bosons and must therefore be included in chiral scalar superfields. We will define U_i , D_i , $\bar{U}_i$, $\bar{D}_i$, N_i , E_i , and $\bar{E}_i$ as the left-chiral superfields whose ψ_L components are the left-handed fields of the quarks of charge $2e/3$ and $-e/3$, of the antiquarks of charge $-2e/3$ and $+e/3$, of the leptons of charge 0 and $-e$, and of the antileptons of charge $+e$, respectively, with i a generation label running over values 1, 2, and 3. (For instance, the spinor components of U_1 , U_2 , and U_3 are the left-handed fields of the u , c , and t quarks, respectively.) Of these superfields, U_i and D_i form $SU(2)$ doublets, N_i and E_i also form $SU(2)$ doublets, and the others are $SU(2)$ singlets. The quark superfields form $SU(3)$ triplets and the antiquark superfields form $SU(3)$ antitriplets, with color indices suppressed, and the leptons and antilepton superfields are $SU(3)$ singlets. As mentioned earlier, the particles described by the scalar components of these superfields are known as squarks, antisquarks, sleptons, and antisleptons. There are also the gauginos, the spin 1/2 superpartners of the gauge bosons of $SU(3)$, $SU(2)$, and $U(1)$, respectively known as the gluino, wino, and bino.¹

We must also add some mechanism that produces a spontaneous breakdown of $SU(2) \cdot U(1)$ and gives mass to all the quarks and leptons as well as to the $W^\pm$ and Z^0 . The simplest possibility is to suppose the existence of just two more $SU(2)$ doublets of left-chiral superfields:

$$H_1 = \begin{pmatrix} H_1^0 \\ H_1^- \end{pmatrix}, \quad H_2 = \begin{pmatrix} H_2^+ \\ H_2^0 \end{pmatrix}, \quad (28.1.1)$$

which appear in the Lagrangian density in linear combinations of the $SU(3) \cdot SU(2) \cdot U(1)$ -invariant $\mathcal{F}$ -terms:

$$[(D_i H_1^0 - U_i H_1^-) \bar{D}_j]_{\mathcal{F}}, \quad [(E_i H_1^0 - N_i H_1^-) \bar{E}_j]_{\mathcal{F}}, \quad (28.1.2)$$

and

$$[(D_i H_2^+ - U_i H_2^0) \bar{U}_j]_{\mathcal{F}}, \quad (28.1.3)$$

¹As discussed in Section 28.3, the energy scale that characterizes supersymmetry breaking is expected to be considerably higher than the ≈ 300 GeV that characterizes the breaking of $SU(2) \cdot U(1)$, so there is a substantial range of energies in which supersymmetry but not $SU(2) \cdot U(1)$ may be considered to be broken. In this range, the gauginos have masses that are governed by $SU(2) \cdot U(1)$ symmetry, so the neutral electroweak gauginos of definite mass are superpartners of the $SU(2)$ triplet W^0 and the $SU(2)$ singlet B , known as the neutral wino and the bino, rather than superpartners of the Z^0 and the photon. When $SU(2) \cdot U(1)$ breaking is taken into account there is a small mixing of the neutral wino and bino.

with obvious contractions of color indices. According to Eq. (26.4.11), a non-vanishing expectation value of the scalar component of H_1^0 gives mass to the charged leptons and charge $-e/3$ quarks, while a non-vanishing expectation value of the scalar component of H_2^0 gives mass to the charge $+2e/3$ quarks. These expectation values of course also give mass to the $W^\pm$ and Z^0 vector bosons and, since the H_1 and H_2 are $SU(2)$ doublets, we automatically get the same successful results for these masses as found in Section 21.3. Note that supersymmetry does not allow the complex conjugates of the H_1 and H_2 left-chiral superfields to appear in the superpotential, so a vacuum expectation value of the scalar component of H_1^0 cannot give masses to the charge $+2e/3$ quarks, and a vacuum expectation value of the scalar component of H_2^0 cannot give mass to the charge $-e/3$ quarks or the charged leptons, which is why both H_1 and H_2 are needed to give masses to all the quarks and leptons.

Of course, there might be more than one of the H_1 and/or H_2 doublets. Their numbers are partly constrained by the condition of anomaly cancellation. We saw in Section 22.4 that the gauge symmetries of the non-supersymmetric standard model are anomaly-free, as they must be for quantum mechanical consistency, but now there are additional spinor fields in the Lagrangian. The gaugino fields don't create any problems, because their left-handed components belong to the adjoint representation of the gauge group, which is real for all gauge groups. The only problem can arise from the higgsinos—the spin 1/2 components of the superfields (H_1^0, H_1^-) and (H_2^+, H_2^0) . The spinor components of each (H_1^0, H_1^-) doublet of superfields produces an $SU(2)$ - $SU(2)$ - $U(1)$ anomaly proportional to $\sum t_{3y}^2 = (\frac{1}{2}g)^2(\frac{1}{2}g') + (-\frac{1}{2}g)^2(\frac{1}{2}g') = \frac{1}{4}g^2g'$, while the spinor components of each (H_2^+, H_2^0) doublet of superfields produces an $SU(2)$ - $SU(2)$ - $U(1)$ anomaly proportional to $\sum t_{3y}^2 = (\frac{1}{2}g)^2(-\frac{1}{2}g') + (-\frac{1}{2}g)^2(-\frac{1}{2}g') = -\frac{1}{4}g^2g'$. *The cancellation of anomalies thus requires an equal number of (H_1^0, H_1^-) and (H_2^+, H_2^0) doublets.* In this case, all anomalies cancel, including the $U(1)^3$ and $U(1)$ -graviton-graviton anomalies. The next section will give an argument that there is in fact just one of each type of doublet.

In a theory constructed along these lines, we have to give up one of the attractive features of the non-supersymmetric standard model, that is, that it *automatically* excludes any renormalizable interactions that violate the conservation of baryon or lepton number. There are several renormalizable supersymmetric $SU(3) \cdot SU(2) \cdot U(1)$ -invariant $\mathcal{F}$ -terms that could be included in the Lagrangian density that would violate baryon and/or lepton number conservation without violating the $SU(3) \cdot SU(2) \cdot U(1)$ gauge symmetries:

$$[(D_i N_j - U_i E_j) \bar{D}_k]_{\mathcal{F}}, \quad [(E_i N_j - N_i E_j) \bar{E}_k]_{\mathcal{F}}, \quad (28.1.4)$$

and also

$$[\bar{D}_i \bar{D}_j \bar{U}_k]_{\mathcal{F}}, \quad (28.1.5)$$

with the three suppressed color indices in Eq. (28.1.5) understood to be contracted with an antisymmetric ϵ -symbol to give a color singlet. With all of these interactions present, there would be no clever way to assign baryon and lepton numbers to the squarks and sleptons that would avoid an unsuppressed violation of baryon and lepton number conservation. For instance, the exchange of the scalar boson of the $\bar{D}$ superfield between vertices for

interactions (28.1.4) and (28.1.5) would lead to the process $u_L d_R u_R \rightarrow \bar{e}_R$, observed for instance as $p \rightarrow \pi^0 + e^+$, at a catastrophic rate that is only suppressed by factors of coupling constants. To avoid this, it is necessary to make an independent assumption that would rule out some or all of the interactions (28.1.4)–(28.1.5).

Note that it is not necessary to rule out *all* of the interactions (28.1.4) and (28.1.5). For instance, suppose that we assume only that baryon number is conserved, with conventional baryon number assignments: the U_i and D_i left-chiral superfields are assigned baryon number $+1/3$; the $\bar{U}_i$ and $\bar{D}_i$ are assigned baryon number $-1/3$; and the L_i , E_i , H_1 , and H_2 all are assigned baryon number 0. This would allow the interactions (28.1.4) while forbidding the interactions (28.1.5). Despite appearances, the interactions (28.1.4) alone do not violate the conservation of lepton number, provided the scalar components of the superfields are assigned appropriate lepton numbers. This can be done by assigning lepton number 0 to the N_i and E_i superfields, lepton number -1 to the U_i , D_i , $\bar{U}_i$, and $\bar{D}_i$ superfields, lepton number -2 to the $\bar{E}_i$ superfields, lepton number 0 to the H_1 and H_2 superfields, and lepton numbers -1 and $+1$ to θ_L and θ_R , respectively. (Recall that such symmetries, under which θ transforms non-trivially, are known as R -symmetries.) Then all quarks and leptons have conventional lepton numbers: the fermion components ν_{iL} and e_{iL} , which are the coefficients of θ_L in N_i and E_i , have the lepton numbers $0 + 1 = +1$; the fermion components $\bar{e}_{iR}$ of the $\bar{E}_i$ superfields have lepton numbers $-2 + 1 = -1$, and the quarks and antiquarks have lepton numbers $-1 + 1 = 0$. The higgsinos (the fermion components of H_1 and H_2) have lepton numbers $0 + 1 = +1$. On the other hand, the scalar components of the superfields have the same lepton numbers as the superfields themselves, which are unconventional. Further, the $\mathcal{F}$ -term of a left-chiral superfield is the coefficient of θ_L^2 , so the interactions in Eq. (28.1.4) have lepton numbers $-1 + 0 - 1 + 2 = 0$ and $0 + 0 - 2 + 2 = 0$; the H_1 interactions (28.1.2) have lepton number $-1 + 0 - 1 + 2 = 0$ and $0 + 0 - 2 + 2 = 0$, respectively; and the H_2 interaction (28.1.3) has lepton number $-1 + 0 - 1 + 2 = 0$; so none of these interactions violate the conservation of lepton number. Also, the scalar components of H_1 and H_2 have lepton number 0, so their vacuum expectation values also do not violate the conservation of lepton number. With this assignment of lepton numbers, lepton number conservation rules out any renormalizable interactions that would violate baryon number conservation: the interaction (28.1.5) has lepton number $-1 - 1 - 1 + 2 = -1$, and so is forbidden.

The interactions (28.1.4) would allow an alternative mechanism for breaking $SU(2) \cdot U(1)$ and giving mass to the charged leptons and charge $-e/3$ quarks: the scalar components of the neutrino superfields N_i might have non-vanishing vacuum expectation values. (With the lepton number assignments of the previous paragraph this expectation value would not violate lepton number conservation, because these scalar components have the lepton number of the N_i superfields, which is zero.) But we cannot rely on this mechanism to let us do without the H_1 superfields altogether, because we still need the H_2 interactions (28.1.3) to give mass to the charge $+2e/3$ quarks and, as we have seen, the cancellation of anomalies requires equal numbers of H_1 and H_2 superfields.

It is usually assumed instead that some symmetries forbid both interactions (28.1.4) and (28.1.5). Obviously, these symmetries could be baryon and lepton number conservation,

with conventional assignments of these numbers: U_i, D_i having baryon number $B = 1/3$ and lepton number $L = 0$, $\bar{U}_i$ and $\bar{D}_i$ having baryon number $B = -1/3$ and lepton number 0, N_i and E_i having lepton number $L = +1$ and baryon number 0, $\bar{E}_i$ having lepton number -1 and baryon number 0, and H_1^0, H_1^- and H_2^+, H_2^0 and θ_L and θ_R all having baryon and lepton numbers 0. The same results apply if we only assume the conservation of certain linear combinations of baryon and lepton number, such as the anomaly-free combination $B - L$ discussed in Section 22.4.

There are widespread doubts about whether it is possible to have exact continuous global symmetries, because in string theory the existence of any exact continuous symmetry would imply the existence of a massless spin 1 particle coupled to the symmetry current, so that the symmetry would have to be local, not global [1b]. But the interactions (28.1.4) and (28.1.5) may also be banned by assuming a *discrete* global symmetry, known as the conservation of R parity [2]. The R parity is defined to be $+1$ for quarks, leptons, gauge bosons, and Higgs scalars, and -1 for their superpartners. This R parity is equal to

$$\Pi_R = (-1)^F (-1)^{3(B-L)}, \quad (28.1.6)$$

where $(-1)^F$ is the fermion parity, which is $+1$ for all bosons and -1 for all fermions. Fermion parity is the same sign as is produced by a 2π rotation, and is therefore always conserved, so if $B - L$ is conserved then so is R parity.² It is possible that R parity may be conserved even if $B - L$ is not, but in fact the interactions (28.1.4) and (28.1.5) are forbidden by R parity conservation, so as far as renormalizable interactions are concerned R parity conservation implies the conservation of both baryon and lepton number. This is *not* true of the non-renormalizable supersymmetric interactions that are presumably produced by physical processes at very high energies. The baryon- and lepton-non-conserving processes produced by such interactions are discussed in Section 28.7.

All of the new “sparticles” (squarks, sleptons, gauginos, and higgsinos) that are required by supersymmetry theories have negative R parity, so if R parity is exact and unbroken then the lightest of the new particles required by supersymmetry must be absolutely stable. All of the other new particles will then undergo a chain of decays, ultimately yielding ordinary particles and the lightest new particle. Much of the phenomenology of various supersymmetry models is governed by the choice of which of the new particles is the lightest.

With supersymmetry and either R parity or $B - L$ conserved, the most general renormalizable Lagrangian for the superfields discussed above consists of the usual gauge-invariant kinematic part for the chiral superfields, given by a sum of terms of the form $(\Phi^* \exp(-V) \Phi)_D$ for each of the quark, lepton, and Higgs chiral superfields, plus the usual gauge-invariant kinematic term for the gauge superfields, given by a sum of terms of the form $\epsilon_{\alpha\beta} (W_\alpha W_\beta)_F$ for each of the $SU(3)$, $SU(2)$, and $U(1)$ field strength superfields, plus the supersymmetric Yukawa couplings, given by a linear combination of the interactions (28.1.2), (28.1.3), and

²The value of $(-1)^{3(B-L)}$ is -1 for quark and lepton superfields and $+1$ for all other superfields, so the conservation of R parity is equivalent to invariance under a transformation in which all quark and lepton superfields change sign with no change in other superfields. This invariance principle was introduced in Reference 3 in order to rule out the interactions (28.1.4) and (28.1.5).

a new $\mathcal{F}$ -term coupling H_1 and H_2 :

$$\begin{aligned} \mathcal{L}_Y = & \sum_{ij} h_{ij}^D [(D_i H_1^0 - U_i H_1^-) \bar{D}_j]_{\mathcal{F}} + \sum_{ij} h_{ij}^E [(E_i H_1^0 - N_i H_1^-) \bar{E}_j]_{\mathcal{F}} \\ & + \sum_{ij} h_{ij}^U [(D_i H_2^+ - U_i H_2^0) \bar{U}_j]_{\mathcal{F}} + \mu [H_2^+ H_1^- - H_2^0 H_1^0]_{\mathcal{F}} + \text{H.c.} \end{aligned} \quad (28.1.7)$$

As we will see in Section 28.3, more terms will have to be added to the Lagrangian in order to account for supersymmetry breaking.

The coefficient μ in Eq. (28.1.7) has the dimensions of mass, and is the only dimensional parameter that enters in the supersymmetric version of the standard model Lagrangian. It is somewhat disappointing to find that this term is still allowed, because it revives the hierarchy problem: why is μ not of order 10^{16} to 10^{18} GeV? The μ -term in Eq. (28.1.7) can be avoided if we assume that lepton number is conserved, with the unconventional lepton number assignments discussed above that would allow the interactions (28.1.4) but not the interactions (28.1.5). In this case, the μ -term carries lepton number $+2$ and is therefore also forbidden. This term can also be forbidden if we assume a $U(1)$ Peccei–Quinn symmetry [4], for which the superfields H_1 and H_2 carry equal quantum numbers, say $+1$, while θ_L and θ_R are neutral. The interactions (28.1.2) and (28.1.3) which give mass to the quarks and leptons are then allowed if, for instance, we give Peccei–Quinn quantum numbers -1 to the left-chiral superfields of antisquarks and antileptons while the left-chiral superfields of squarks and sleptons are taken to be neutral. This choice then also forbids the dangerous interactions (28.1.4) and (28.1.5). Unfortunately, as we will see in Section 28.4, the μ -term in Eq. (28.1.7) seems to be needed for phenomenological reasons. The theories of gravity-mediated supersymmetry breaking discussed in Section 31.7 provide a natural mechanism for producing a μ -term of an acceptable magnitude.

We can obtain a crude upper bound on the masses of the new particles by assuming that supersymmetry does solve the hierarchy problem discussed at the beginning of this chapter. In accord with the theorem of Section 27.6, if supersymmetry were unbroken the contribution to the mass of the scalar component of H_1 or H_2 from one-loop diagrams with an intermediate quark, lepton, W or Z loop would be cancelled by the corresponding one-loop diagram with an intermediate squark, slepton, wino, or bino. Therefore with supersymmetry broken the contribution δm_H^2 of such diagrams to the squared masses of the H_1 and H_2 scalars is a sum of terms of order $(\mathcal{G}_s^2/8\pi^2)\Delta m_s^2$, where $\mathcal{G}_s$ is the Yukawa or gauge coupling of the Higgs scalar to supermultiplet s , and Δm_s^2 is the mass-squared splitting within the supermultiplet. To avoid having to fine-tune these corrections, we need δm_H^2 not much larger than the coefficient of order $(300 \text{ GeV})^2$ of the term in the standard model Lagrangian density that gives the observed $SU(2) \cdot U(1)$ breaking in the tree approximation, so we will assume that $\delta m_H^2 < (1 \text{ TeV})^2$. For instance, the top quark and squark have couplings to H_2 of order unity, so we expect that the splittings Δm^2 should be less than about $8\pi^2 \text{ TeV}^2$, and hence the masses of the top squarks should be less than about 10 TeV. We will see in Section 28.4 that the rates of flavor-changing processes can be brought within experimental upper bounds by taking the masses of the squarks to be nearly equal, in which case this can be taken as a rough upper limit on the masses of all the

squarks. (However it is possible that the rates of these processes may be suppressed instead by very large masses of the first two generations of squarks, while the mass of the top squark is below the 10 TeV naturalness bound [4a].) The limits set by this sort of argument on the masses of other particles with $R = -1$ are somewhat weaker, but at least in the popular class of models discussed in Section 28.6, none of the masses of these particles are expected to be much greater than the squark masses, so 10 TeV can be taken as an upper bound on all of them. On the other hand, the fact that none of these particles have been observed only indicates that their masses are probably greater than about 100 GeV, so there is an ample mass range in which they may yet be found.

* * *

If R parity conservation or some other conservation law makes the lightest of the new particles predicted by supersymmetry stable, then some of these particles may be left from the early universe. The number density of these relics can be estimated using techniques that were originally applied to the cosmic density of massive neutrinos [4b]. To give one example of this sort of calculation, we shall show that for a broad range of plausible masses, the new stable particle of supersymmetry theories cannot be a charged and uncolored particle, like a charged slepton, wino, or higgsino [4c].

Once the cosmic temperature T (in energy units, with the Boltzmann constant set equal to unity) drops below the mass m of any stable charged untrapped particle, their number nR^3 in a volume R^3 that expands with the universe is decreased by annihilation at a rate per particle equal to $\overline{v\sigma}n$, where $\overline{v\sigma}$ is the mean value of the product of the relative velocity and the annihilation cross-section. That is,

$$\frac{d(nR^3)}{dt} = -\overline{v\sigma} n^2 R^3,$$

so that

$$\frac{1}{nR^3} = \left(\frac{1}{nR^3} \right)_0 + \int_{t_0}^t \frac{\overline{v\sigma}}{R^3} dt, \quad (28.1.8)$$

where 0 labels the epoch at which $T \simeq m$. The annihilation process is exothermic, so $\overline{v\sigma}$ approaches a constant for $v \ll 1$. Also, in a radiation-dominated phase of the cosmic expansion $R \propto t^{1/2}$, so the integral converges, and gives

$$\begin{aligned} \left(\frac{1}{nR^3} \right)_{t \rightarrow \infty} &= \left(\frac{1}{nR^3} \right)_0 \overline{v\sigma} \int_{t_0}^{\infty} \frac{dt}{R_0^3 (t/t_0)^{3/2}} \\ &= \left(\frac{1}{nR^3} \right)_0 \frac{2\overline{v\sigma}t_0}{R_0^3}. \end{aligned} \quad (28.1.9)$$

The density n_B of baryon number (baryons minus antibaryons) goes as R^{-3} , so this can be rewritten as a formula for the present ratio of new particles to baryons:

$$(n/n_B)_{\infty} = [(n_B/n)_0 + 2\overline{v\sigma} n_{B0}t_0]^{-1}. \quad (28.1.10)$$

We expect that the ratio $(n/n_B)_0$ at the time that T drops to a value $\approx m$ is roughly of order unity, and since in any realistic theory the present ratio $(n/n_B)_{\infty}$ must be much

less than unity, we can neglect the first term in the denominator on the right-hand side of Eq. (28.1.10), and write instead

$$(n/n_B)_\infty \simeq \frac{1}{\bar{v}\sigma n_{B0}t_0}. \quad (28.1.11)$$

The precise value of $\bar{v}\sigma$ depends on the particle spin and its interactions; keeping track only of factors of 2π , the particle mass m , and the electric charge, we can estimate it generally to be of order

$$\bar{v}\sigma \approx \frac{e^4\mathcal{N}}{2\pi m^2} \approx 10^{-3} \frac{\mathcal{N}}{m^2}, \quad (28.1.12)$$

where $\mathcal{N}$ is the number of charged particle spin states with mass less than m , into which this particle may annihilate. Also, the age of the universe at a temperature $T_0 \simeq m$ is $t_0 \simeq m_{\text{PL}}/m^2$, where $m_{\text{PL}} \simeq 10^{18}$ GeV, and the density of baryon number is about 10^{-9} times the photon number density, which is of order T^3 , so that $n_{B0} \simeq 10^{-9}m^3$. Putting this together, we find the present ratio of the new charged particles to baryons:

$$(n/n_B)_\infty \approx 10^{12} \frac{m}{m_{\text{PL}}\mathcal{N}} \approx 10^{-6} \frac{m(\text{GeV})}{\mathcal{N}}. \quad (28.1.13)$$

These new charged particles would experience the same condensations into galaxies, stars, and planets as ordinary baryons, so this would be the ratio observed today on earth. But experiments [4d] that apply mass spectroscopy to samples of water that have been strongly enriched in heavy water-like molecules by electrolysis have set limits of about $10^{-21}n_B$ on the number density of new charged particles with $6 \text{ GeV} < m < 330 \text{ GeV}$ in terrestrial matter. Thus, even if $\mathcal{N}$ is as large as 1000, these measurements decisively rule out the existence of any new charged untrapped particles in this mass range in the numbers that would have been left over from the early universe.

On the other hand, neutral untrapped particles would be left in intergalactic space. Such particles might well provide the “missing mass,” that seems to be necessary to account for the gravitational field that governs the motion of galaxies in clusters of galaxies. One of these possible neutral particles is the gravitino, whose cosmological abundance is discussed in Section 28.3. Ellis *et al.* [4c] have extended cosmological considerations to all the new particles required by supersymmetry.

28.2 Supersymmetry and Strong–Electroweak Unification

We shall have to defer a detailed assessment of supersymmetric models of particle physics until we are ready to consider how supersymmetry is broken. In this section we will consider the quantitative application of supersymmetry in one context in which the mechanism for the breakdown of supersymmetry is relatively unimportant, and in which supersymmetry has scored what so far is its greatest empirical success.

If the $SU(3) \cdot SU(2) \cdot U(1)$ gauge group of the strong and electroweak interactions is embedded in a simple group G that has the known quarks and leptons (plus perhaps some $SU(3) \cdot SU(2) \cdot U(1)$ -neutral fermions) as a representation, then, as described in

Section 21.5, at energies at or above the scale M_X at which G is spontaneously broken, the $SU(3) \cdot SU(2) \cdot U(1)$ coupling constants will be related by

$$g_s^2 = g^2 = \frac{5g'^2}{3} \quad \text{at energies} \geq M_X. \quad (28.2.1)$$

At energies far below M_X , these couplings are seriously affected by renormalization corrections. If measured at a scale $\mu < M_X$, the couplings will have values $g'^2(\mu)$, $g^2(\mu)$, $g_s^2(\mu)$, governed by one-loop renormalization group equations

$$\mu \frac{d}{d\mu} g'(\mu) = \beta_1(g'(\mu)), \quad \mu \frac{d}{d\mu} g(\mu) = \beta_2(g(\mu)), \quad \mu \frac{d}{d\mu} g_s(\mu) = \beta_3(g_s(\mu)), \quad (28.2.2)$$

with initial conditions at M_X satisfying Eq. (28.2.1). In the original use [5] of these renormalization group equations, discussed in Section 21.5, the beta functions were calculated in one-loop order to be

$$\beta_1 = \frac{5n_g g'^3}{36\pi^2}, \quad (28.2.3)$$

$$\beta_2 = \frac{g^3}{4\pi^2} \left(-\frac{11}{6} + \frac{n_g}{3} \right), \quad (28.2.4)$$

$$\beta_3 = \frac{g_s^3}{4\pi^2} \left(-\frac{11}{4} + \frac{n_g}{3} \right), \quad (28.2.5)$$

where n_g is the number of generations of quarks and leptons and the relatively small contributions of scalar fields are here neglected. Since M_X will turn out to be many orders of magnitude larger than the energies accessible with today's accelerators, it seems reasonable to suppose that supersymmetry is unbroken over most of the range below M_X , in which case all of the new fields discussed in the previous section need to be included in calculations of the beta functions in Eq. (28.2.1). These new fields introduce three major changes in the calculations of the beta functions:

1. For every gauge boson, there is a Majorana gaugino with the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers. Equation (17.5.41) shows that the ratio of the contribution to the beta function for any gauge coupling of a Dirac fermion that furnishes a representation of the gauge group with generators t_A to the contribution of the corresponding gauge boson is $-4C_2/11C_1$, where according to Eqs. (17.5.33) and (17.5.34) the ratio of C_1 and C_2 is given by:

$$\sum_{AB} C_{CAB} C_{DBA} = -(C_1/C_2) \text{Tr}(t_C t_D). \quad (28.2.6)$$

For the adjoint representation, $(t_C)_{AB} = iC_{ABC}$, so $C_1 = C_2$, and so a Dirac fermion in the adjoint representation makes a contribution that is $-4/11$ that of the gauge bosons. But the gauginos are Majorana fermions, so their contribution is $-2/11$ that of the gauge boson. Thus the term $11/6$ and $11/4$ in Eqs. (28.2.4) and (28.2.5) are reduced by a factor $9/11$ to $9/6$ and $9/4$, respectively.

2. For every left-handed quark, lepton, antiquark, or antilepton field, there is a complex scalar field with the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers. Following the same method as in Section 17.5, it is not hard to calculate that the contribution of a complex

scalar field belonging to a representation of a gauge group with generators t_A to the beta function for the gauge coupling g_i is

$$[\beta_i(g_i)]_{\text{scalar}} = \frac{g_i^3 C_{2i}}{48\pi^2}, \quad (28.2.7)$$

where $\text{Tr}(t_A t_B) = g_i^2 C_{2i} \delta_{AB}$. This is $1/4$ the contribution of a Dirac spinor field in the same representation, given by Eq. (18.7.2), and hence $1/2$ the contribution of each left-handed spinor field (including the complex conjugates of the right-handed components of the Dirac fields). Thus the coefficients of n_g in Eqs. (28.2.3)–(28.2.5) should be increased by a factor $3/2$.

3. The decrease by a factor $9/11$ of the negative gauge boson terms in the beta function and the increase by a factor $3/2$ of the positive squark and slepton terms both lead to a general decrease in the rate at which the three gauge coupling constants diverge below M_X from the ratios (28.2.1). This will increase our estimate of M_X , but, as we shall see, in itself it would have no affect on the prediction of the electroweak mixing parameter $\sin^2 \theta$. But these changes do enhance the relative contribution of the Higgs scalars, which was neglected in Eqs. (28.2.3)–(28.2.5), and which is now also accompanied with the larger contribution of the accompanying higgsinos. With n_s of the superfields (H_1^0, H_1^-) or (H_2^+, H_2^0) discussed in the previous section, the constant C_{2i} in Eq. (28.2.7) is $[(1/2)^2 + (-1/2)^2]n_s = n_s/2$ for $SU(2)$, and is $2n_s(\pm 1/2)^2 = n_s/2$ for $U(1)$. According to Eq. (28.2.7), the scalar components of these superfields make a contribution to β_1 equal to $n_s g^3/96\pi^2$ and a contribution to β_2 also equal to $n_s g^3/96\pi^2$. As we have seen, the Majorana higgsinos contribute twice as much to the beta functions as complex scalars with the same quantum numbers, so the superfields (H_1^0, H_1^-) or (H_2^+, H_2^0) make a total contribution to β_1 and β_2 that is $3/2$ the contribution of the Higgs scalars, and hence equal to $n_s g^3/32\pi^2$ and $n_s g^3/32\pi^2$, respectively.

Making all these changes in the beta functions, we now have

$$\beta_1 = \frac{g^3}{4\pi^2} \left(\frac{5n_g}{6} + \frac{n_s}{8} \right), \quad (28.2.8)$$

$$\beta_2 = \frac{g^3}{4\pi^2} \left(-\frac{9}{6} + \frac{n_g}{2} + \frac{n_s}{8} \right), \quad (28.2.9)$$

$$\beta_3 = \frac{g_s^3}{4\pi^2} \left(-\frac{9}{4} + \frac{n_g}{2} \right). \quad (28.2.10)$$

The solutions of the renormalization group equations (28.2.2) are then

$$\frac{1}{g'^2(\mu)} = \frac{1}{g'^2(M_X)} + \frac{1}{2\pi^2} \left(\frac{5n_g}{6} + \frac{n_s}{8} \right) \ln \left(\frac{M_X}{\mu} \right), \quad (28.2.11)$$

$$\frac{1}{g^2(\mu)} = \frac{1}{g^2(M_X)} + \frac{1}{2\pi^2} \left(-\frac{3}{2} + \frac{n_g}{2} + \frac{n_s}{8} \right) \ln \left(\frac{M_X}{\mu} \right), \quad (28.2.12)$$

$$\frac{1}{g_s^2(\mu)} = \frac{1}{g_s^2(M_X)} + \frac{1}{2\pi^2} \left(-\frac{9}{4} + \frac{n_g}{2} \right) \ln \left(\frac{M_X}{\mu} \right). \quad (28.2.13)$$

It is convenient to take $\mu = m_Z$, so that $SU(2) \cdot U(1)$ can be regarded as unbroken over almost all of the range of energies in which we use the formulas (28.2.11)–(28.2.13). Using Eq. (28.2.1), the difference between Eqs. (28.2.12) and (28.2.13) gives

$$\frac{1}{g^2(m_Z)} - \frac{1}{g_s^2(m_Z)} = \frac{1}{2\pi^2} \left(\frac{3}{4} + \frac{n_s}{8} \right) \ln \left(\frac{M_X}{m_Z} \right), \quad (28.2.14)$$

Table 28.1. Values of the electroweak mixing parameter $\sin^2 \theta$ and unification mass M_X given by Eqs. (28.2.17) and (28.2.18), as functions of the number n_s of left-chiral superfield doublets (H_1^0, H_1^-) or (H_2^+, H_2^0) .

n_s	$\sin^2 \theta$	M_X (GeV)
0	0.203	$8.7 \cdot 10^{17}$
2	0.231	$2.2 \cdot 10^{16}$
4	0.253	$1.1 \cdot 10^{15}$

while the difference between Eq. (28.2.12) and 3/5 of Eq. (28.2.11) gives

$$\frac{1}{g^2(m_Z)} - \frac{3}{5g'^2(m_Z)} = \frac{1}{2\pi^2} \left(-\frac{3}{2} + \frac{n_s}{20} \right) \ln \left(\frac{M_X}{m_Z} \right). \quad (28.2.15)$$

Equation (21.3.19) allows us to express the electroweak couplings in terms of the electroweak mixing angle θ and the positron charge e :

$$g(m_Z) = -e(m_Z)/\sin \theta, \quad g'(m_Z) = -e(m_Z)/\cos \theta. \quad (28.2.16)$$

We can then solve for the unknowns $\ln(M_X/m_Z)$ and $\sin^2 \theta$ in terms of input parameters $e(m_Z)$ and $g_s(m_Z)$:

$$\sin^2 \theta = \frac{18 + 3n_s + (e^2(m_Z)/g_s^2(m_Z))(60 - 2n_s)}{108 + 6n_s}, \quad (28.2.17)$$

$$\ln \left(\frac{M_X}{m_Z} \right) = \left(\frac{8\pi^2}{e^2(m_Z)} \right) \left(\frac{1 - (8e^2(m_Z)/3g_s^2(m_Z))}{18 + n_s} \right). \quad (28.2.18)$$

For $n_s = 0$ Eq. (28.2.17) gives the same result (21.5.15) for $\sin^2 \theta$ as was originally calculated (ignoring the small contribution of Higgs scalars) in non-supersymmetric theories, but the value (28.2.18) of $\ln(M_X/m_Z)$ is larger than the original result (21.5.16) by a factor 11/9, which as we have seen arises from gaugino contributions to the beta functions.

Using the same input parameters $e^2(m_Z)/4\pi = (128)^{-1}$, $g_s^2(m_Z)/4\pi = 0.118$, $m_Z = 91.19$ GeV as in Section 21.5 now gives the numerical results shown in Table 28.1. As discussed in the previous section, the necessity of cancelling anomalies in the electroweak currents requires equal numbers of (H_1^0, H_1^-) and (H_2^+, H_2^0) doublets, so we consider only even values of the number n_s of these superfields.

Remarkably, the value $n_s = 2$ for the simplest plausible theory yields a value [6] $\sin^2 \theta = 0.231$ which is in perfect agreement with the experimentally observed value, $\sin^2 \theta = 0.23$.

The value of M_X is 20 times greater [7] than calculated in this way in non-supersymmetric theories, leading to a decrease by a factor 20^{-4} in the rate for proton decay processes like $p \rightarrow \pi^0 + e^+$, thus removing a conflict with the experimental non-observation of such processes. (Proton decay is discussed in more detail in Section 28.7.) This increase in the value of M_X brings it closer to the energy scale $\approx 10^{18}$ GeV at which gravitation has the same strength as other interactions. It may be that this remaining gap may be filled by a change in gravitational interactions at very high energies [7a].

A value $n_s = 4$ would give a value for $\sin^2 \theta$ in serious disagreement with experiment, and a value of M_X low enough to revive the conflict with expectations for proton decay. This makes a strong case for having just one of each superfield (H_1^0, H_1^-) and (H_2^+, H_2^0) .

Unlike the calculated values of $\sin^2 \theta$ and M_X , the calculated value of the common gauge coupling (28.2.1) at M_X does depend on the number of generations as well as the number of scalar doublets. With $n_g = 3$ and $n_s = 2$ and our previous input parameters, Eq. (28.2.13) gives

$$\frac{g^2(M_X)}{4\pi} = \frac{g_s^2(M_X)}{4\pi} = \frac{1}{17.5}. \quad (28.2.19)$$

28.3 Where is Supersymmetry Broken?

Supersymmetry if valid at all is certainly not apparent in the menu of known particles, so any consideration of the implications of supersymmetry at ordinary energies requires us to make some assumption about the mechanism of supersymmetry breaking. It would be simplest to suppose that supersymmetry is broken like $SU(2) \cdot U(1)$, by effects occurring in the tree approximation of the supersymmetric standard model. This possibility may be definitely ruled out.

One argument against a tree-approximation breakdown of supersymmetry is based on the mass sum rule (27.5.11), which holds separately for each value of the unbroken conserved quantities color and electric charge. In the color-triplet sector with electric charge $-e/3$ the only known fermions are the d , s , and b quarks, for which

$$m_d^2 + m_s^2 + m_b^2 \simeq (5 \text{ GeV})^2. \quad (28.3.1)$$

According to the sum rule, if there are no other fermions with this color and charge, then the sum of all squared masses for bosons (counting each spin state separately) with the same color and charge must equal about $2(5 \text{ GeV})^2$. In particular, each of the squarks with this color and charge must have a mass no greater than 7 GeV. The existence of such light squarks is definitely ruled out experimentally; they would have shown up, for instance, as a contribution to the rate for electron-positron annihilation into hadrons at energies where this process has been studied very thoroughly.

This argument could be invalidated if there were a heavy fourth generation of quarks. There is another argument, due to Dimopoulos and Georgi [3], which would apply however many heavy quarks there are and which yields an even stronger upper bound on the mass of the lightest squark. The unbroken conservation of charge and color tells us that the only non-zero D_{A0} -terms in the supersymmetric standard model are for the generators y of $U(1)$ and t_3 of $SU(2)$, which we shall call D_1 and D_2 , respectively. The values of these generators

are $y = -g'/6$ and $t_3 = +g/2$ for the left-handed quarks of charge $2e/3$; $y = -g'/6$ and $t_3 = -g/2$ for the left-handed quarks of charge $-e/3$; $y = 2g'/3$ and $t_3 = 0$ for the right-handed quarks of charge $2e/3$; and $y = -g'/3$ and $t_3 = 0$ for the right-handed quarks of charge $-e/3$. Also, the squark fields are color-triplets, and therefore cannot have vacuum expectation values. According to Eq. (27.5.4), the mass-squared matrix of the charge $2e/3$ color-triplet (not antitriplet) squarks is

$$M_{0U}^2 = \begin{bmatrix} \mathcal{M}_U^* \mathcal{M}_U - g' D_1/6 + g D_2/2 & \mathcal{F}_U^* \\ \mathcal{F}_U & \mathcal{M}_U \mathcal{M}_U^* + 2g' D_1/3 \end{bmatrix}, \quad (28.3.2)$$

while the mass-squared matrix of the charge $-e/3$ color-triplet squarks is

$$M_{0D}^2 = \begin{bmatrix} \mathcal{M}_D^* \mathcal{M}_D - g' D_1/6 - g D_2/2 & \mathcal{F}_D^* \\ \mathcal{F}_D & \mathcal{M}_D \mathcal{M}_D^* - g' D_1/3 \end{bmatrix}. \quad (28.3.3)$$

Also, Eq. (27.5.6) gives the mass-squared matrices of the quarks of charge $2e/3$ and $-e/3$ here as just $\mathcal{M}_U^* \mathcal{M}_U$ and $\mathcal{M}_D^* \mathcal{M}_D$, respectively, with no mixing with the gauginos.

Now let v_u and v_d be the normalized eigenvectors of the quark mass-squared matrices $\mathcal{M}_U^* \mathcal{M}_U$ and $\mathcal{M}_D^* \mathcal{M}_D$ corresponding to the quarks u and d of lowest mass, and consider the expectation values of the corresponding squark mass-squared matrices

$$\begin{bmatrix} 0 \\ v_u^* \end{bmatrix}^\dagger M_{0U}^2 \begin{bmatrix} 0 \\ v_u^* \end{bmatrix} = m_u^2 + \frac{2g' D_1}{3}, \quad (28.3.4)$$

$$\begin{bmatrix} 0 \\ v_d^* \end{bmatrix}^\dagger M_{0D}^2 \begin{bmatrix} 0 \\ v_d^* \end{bmatrix} = m_d^2 - \frac{g' D_1}{3}. \quad (28.3.5)$$

These expectation values are weighted averages of the squared masses of the squarks of charge $2e/3$ and $-e/3$, respectively, so at least one squark of charge $2e/3$ must have a squared mass less than $m_u^2 + 2g' D_1/3$, and at least one squark of charge $-e/3$ must have a squared mass less than $m_d^2 - g' D_1/3$. Thus, depending on the sign of D_1 , *there must be either a squark of charge $2e/3$ lighter than the u quark, or a squark of charge $-e/3$ lighter than the d quark.*

Needless to say, the existence of a charged color-triplet scalar this light would radically change strong-interaction phenomenology. Like the u and d quarks, this colored scalar would appear as an ingredient of hadrons with a ‘constituent’ mass of a few hundred MeV, which is certainly not seen. Since this scalar is electrically charged, it would also be created in pairs in $e^+ - e^-$ annihilation at energies above a few hundred MeV, making a contribution to the annihilation cross-section that would destroy the excellent agreement between theory and experiment for this cross-section. Even worse, since the u and d quarks are so light, and D_1 is expected to be of the order of the supersymmetry-breaking scale, Eqs. (28.3.4) and (28.3.5) indicate that one of the squarks would have a negative squared mass, meaning that this squark field would have to develop a non-vanishing expectation value, breaking both color and charge conservation. We are forced to reject the simple picture of supersymmetry

broken spontaneously in the tree approximation in a supersymmetric version of the standard model.

One way out of this conclusion would be to add another $U(1)$ gauge superfield to the theory. If all the quark superfields carry the same value $\tilde{g}$ of this new $U(1)$ generator, then the corresponding D -term $\tilde{D}$ would make an additive contribution $\tilde{g}\tilde{D}$ to the right-hand sides of both Eqs. (28.3.4) and (28.3.5). If this term were sufficiently large, then it could give a large positive value to all of the squark squared masses, avoiding all the problems mentioned above. But there is no sign of such a new neutral gauge boson at accessible energies, and in any case we would still have an upper bound of 7 GeV on the masses of all the squarks of charge $-e/3$.

It is not necessarily a bad thing that we have to look for the breaking of supersymmetry elsewhere than in the tree approximation of the supersymmetric standard model. If supersymmetry were broken in this approximation, then the characteristic mass that sets the scale of supersymmetry breaking would be some mass parameter in the Lagrangian, which would in turn set the scale of all other masses in the standard model. We would then still be confronted with the hierarchy problem: why is this mass scale so much less than 10^{16} – 10^{18} GeV?

There is one known way to explain such large mass ratios. If supersymmetry is not spontaneously broken in the tree approximation in whatever field theory unifies all the interactions at some high mass scale M_X , then as shown in Section 27.6, it will not be broken in any order of perturbation theory. But it can be broken by non-perturbative effects. In particular, if there is some gauge field with an asymptotically free gauge coupling $\mathcal{G}(\mu)$ at renormalization scale μ , and if $\mathcal{G}^2(\mu)/8\pi^2$ is substantially less than unity for $\mu \approx M_X$, then as discussed in Section 18.3, this gauge interaction will become strong at an energy of order $M_S = M_X \exp(-8\pi^2 b/\mathcal{G}^2(M_X))$, where b is a number of order unity. It is not necessary for $\mathcal{G}^2(M_X)/8\pi^2$ to be very small in order to have M_S many orders of magnitude less than M_X . We will see in Section 29.4 that supersymmetry can indeed be broken in just this way, by a gauge coupling that has become strong at some energy $M_S \ll M_X$. Indeed, this is just what happens to chiral symmetry in quantum chromodynamics; there is no mystery why the proton mass (or at least its main part, due to the dynamical breakdown of chiral symmetry, and not to the tiny masses of the u and d quarks) is so much smaller than the unification scale M_X . Alternatively, the forces that are strong at energy M_S may produce a potential for scalar fields, whose vacuum expectation value then breaks supersymmetry.

There are no signs of any new strong interaction of the known quarks and leptons, so we have to assume that the observed particles of the standard model are neutral with respect to the strong force that breaks supersymmetry. Supersymmetry breaking therefore occurs in a ‘hidden sector’ of particles that do feel this new strong force. The remaining question, then, is what is the mechanism by which supersymmetry breaking in this hidden sector is communicated to the known particles of the standard model? As we shall see, most of our expectations for the phenomenological implications of supersymmetry depend on the answer to this question, rather than on the details of the breakdown of supersymmetry itself.

Of course, the mechanism for communicating supersymmetry breaking to observed par-

ticles must be some sort of interaction that is felt by these particles. There are two leading candidates. One mechanism is provided by the $SU(3) \cdot SU(2) \cdot U(1)$ gauge interactions themselves, to be discussed in Section 28.6. The other is gravitation, or rather the auxiliary fields that are superpartners of the gravitational field, to be discussed in Sections 31.4 and 31.7.

Without going into details here, we can make a crude estimate of the supersymmetry-breaking scale M_S for these two possibilities. For gauge-mediated supersymmetry breaking, we expect that the mass splitting between the observed quarks, leptons, and gauge bosons and their superpartners would be of order $g_s^2/16\pi^2$ or $g'^2/16\pi^2$ or $g^2/16\pi^2$ (where g_s , g , and g' are the $SU(3)$, $SU(2)$, and $U(1)$ gauge couplings), depending on which quantum numbers are carried by the supermultiplet in question. (This guess is verified in Section 28.6.) Hence if the squarks, sleptons, and gauginos have masses in the range of 100 GeV to 10 TeV, as argued at the end of Section 28.1, then the supersymmetry-breaking scale M_S would be higher by two or three orders of magnitude—say, of order 100 TeV. On the other hand, if it is gravitation that serves as the mediator of supersymmetry breaking, then on dimensional grounds we would expect the mass splittings Δm between the observed particles and their superpartners to be of order $\sqrt{G}M_S^2$, or perhaps of order GM_S^3 . (Results of both sorts will be encountered in models described in Section 31.7.) If the squarks, sleptons, and gauginos have masses in the range of 100 GeV to 10 TeV, then M_S would be of order 10^{11} GeV for $\Delta m \approx \sqrt{G}M_S^2$, or 10^{13} GeV for $\Delta m \approx GM_S^3$.

The large difference in estimates of the supersymmetry-breaking scale M_S for gauge- and gravitation-mediated supersymmetry breaking makes an important difference in particle phenomenology and cosmology. As already mentioned several times, supersymmetry dictates that the graviton must have a partner of spin 3/2, the gravitino. When supersymmetry is spontaneously broken at a scale M_S , the gravitino acquires a mass m_g of order $\sqrt{G}M_S^2$. (A precise formula will be given in Section 31.3.) For gauge-mediated supersymmetry breaking, this is very small; if $M_S \approx 100$ TeV then $m_g \approx 1$ eV, so the gravitino would be by far the lightest of the new particles required by supersymmetry—that is, the lightest particle with negative R parity (28.1.6). On the other hand, for gravitationally mediated supersymmetry breaking the gravitino mass is just of the same order of magnitude $\sqrt{G}M_S^2$ as the mass splitting between known particles and their superpartners, so the gravitino would have roughly the same mass as the squarks, sleptons, and gauginos. The gravitino then might or might not be the lightest particle with negative R parity, but its interactions with known particles and their superpartners in this case are of gravitational strength, so that gravitinos would play no direct role in experiments on elementary particles.

* * *

There are limits on the number of gravitinos that could survive from the big bang, which set useful constraints on the scale M_S of supersymmetry breaking. At some point in the distant past the temperature T was presumably high enough so that even purely gravitational interactions would have kept gravitinos in thermal equilibrium with other particles, in which case the number density of gravitinos would have been of the order of

T^3 , roughly the same as the number density of photons. (We are using units in which the Boltzmann constant k_B as well as $\hbar$ and c are equal to unity.) If gravitinos do not annihilate or decay, then the expansion of the universe will lower their number density in the same way as the number density of photons, so even after the gravitinos go out of equilibrium they would be present in numbers comparable to those of photons. More precisely, since the photons but not the gravitinos are heated by the annihilation of other particles, the number density n_{g0} of gravitinos at present would be one or two orders of magnitude less than the number density $n_{\gamma 0}$ of photons in the cosmic microwave radiation background. In order for the mass density $m_g n_{g0}$ of the gravitinos not to exceed the upper bound on the cosmic mass density set by the observed value of the Hubble constant, m_g would have to be less [8] than about 1 keV. As we have seen, this limit is well satisfied in theories of gauge-mediated supersymmetry breaking, where the gravitino is too light for cosmic gravitinos to contribute appreciably to the mass density of the universe. Since some of the fields that break supersymmetry in these theories must interact at least indirectly with the known quark, lepton, and gauge fields in order for the known particles to show the effects of supersymmetry breaking, the interactions of the gravitino with the known particles and their superpartners are suppressed only by powers of gauge and Yukawa coupling constants, so all the superpartners of the quarks, leptons, and gauge bosons would decay quickly into these known particles and gravitinos. Thus these particles also do not provide candidates in these models for the ‘missing mass’ sought by cosmologists. (It is possible that conservation laws could keep some particles of the supersymmetry-breaking sector stable, in which case they could conceivably serve as the missing mass.)

On the other hand, for supersymmetry breaking that is gravitationally mediated, gravitinos are heavy enough to be unstable (though gravitino annihilation is still negligible), so the above limit need not apply [9]. We will see in Section 31.3 that the coupling of the gravitino to other fields is proportional to $\sqrt{G}$, so on dimensional grounds the decay rate Γ_g of a gravitino at rest is roughly of the order of Gm_g^3 . This is to be compared with the rate of expansion of the universe, which at temperature T is of order $\sqrt{G}T^4$. (We are here ignoring factors of order 10–100, including those involving non-gravitational coupling constants and the number of particle species.) When the cosmic temperature drops to the value $T \approx m_g$ at which gravitinos become non-relativistic, the ratio of their decay rate to the expansion rate is of order $\sqrt{G}m_g = m_g/m_{\text{Planck}} \ll 1$, so gravitino decay becomes significant only after this time, when the gravitinos are highly non-relativistic. As we have seen, their number density will be of order T^3 , so their energy density will then be of order $m_g T^3$, which is greater than the energy density of order T^4 of the photons and other particles in thermal equilibrium at temperature T , and therefore makes the dominant contribution to the cosmic gravitational field that governs the rate of expansion of the universe. The expansion rate under these conditions is therefore of order $\sqrt{Gm_g T^3}$, and gravitino decay becomes significant when this equals the gravitino decay rate of order Gm_g^3 , and therefore at a temperature

$$T_g \approx G^{1/3} m_g^{5/3}.$$

As we have seen, if these gravitinos did not decay before the present then their mass

had better be less than 1 keV, but they can lead to cosmological difficulties even if they did decay before now. After they decay, their energy must go into the energy of photons and other relativistic particles, so the temperature T'_g after decay is related to the temperature T_g calculated above by the energy conservation condition $m_g T_g^3 \approx T'^4_g$, and hence

$$T'_g \approx G^{1/4} m_g^{3/2}.$$

In particular, since $T_g \ll m_g$, we have $T'_g \gg T_g$. If T_g were less than the temperature $T_n \simeq 0.1$ MeV at which cosmological nucleosynthesis can occur, then gravitinos would still be abundant before nucleosynthesis, giving a higher energy density and hence a faster expansion, so that there would be less time for free neutrons to decay before being incorporated into complex nuclei, and hence more helium would be produced when nucleosynthesis occurs. Also, the ratio of the photon and baryon densities would have been subsequently increased by gravitino decay, so this ratio at the time of nucleosynthesis would have been considerably less than is usually estimated from the present cosmic microwave background temperature, and so nuclear reactions would have incorporated neutrons more completely into helium, and less deuterium would be left today. The present agreement between theory and observation for the cosmic helium and deuterium abundances would thus be destroyed. This problem is avoided if $T_g > 0.1$ MeV, but it can also be avoided under the much weaker condition that $T'_g > 0.4$ MeV, because then after the gravitinos decay the temperature would have been high enough to break up the excess helium and give cosmological nucleosynthesis a fresh start as the universe recools. This condition requires that $m_g > 10$ TeV, which is just barely consistent with the upper bound derived in Section 28.1 on the masses of the superpartners of the known quarks, leptons, and gauge bosons, which for gravitationally mediated supersymmetry breaking are of order m_g . This limit on m_g corresponds to a supersymmetry-breaking scale $M_S > 10^{11}$ GeV for $m_g \approx \sqrt{G} M_S^2$ or $M_S > 10^{13}$ GeV for $m_g \approx G M_S^3$.

28.4 The Minimal Supersymmetric Standard Model

In the previous section we identified two different ways that the breakdown of supersymmetry at a high energy scale M_S could be communicated to the known quarks and leptons: through gauge or gravitational superfields. The supersymmetry-breaking terms in the resulting low-energy effective Lagrangian will then be suppressed by powers of gauge couplings or the Newton constant. Most of these terms will therefore be rather small, with the exception that, along with factors of gauge couplings or the Newton constant, on dimensional grounds the mass terms and other superrenormalizable terms in the effective Lagrangian will be proportional to one or more factors of the supersymmetry-breaking scale M_S , which is quite large compared with known particle masses. We can conclude then that, to a fair approximation for gauge-mediated supersymmetry breaking and a very good approximation for gravitationally mediated supersymmetry breaking, the main effect of supersymmetry breaking will be in the superrenormalizable terms of the effective Lagrangian of the supersymmetric standard model. This version of the standard model [10], which is supersymmetric except for superrenormalizable terms, is usually known as the *minimal supersymmetric standard model*.

With R parity or $B - L$ conserved, the most general superrenormalizable Lagrangian density allowed by $SU(3) \cdot SU(2) \cdot U(1)$ gauge symmetry takes the form

$$\begin{aligned}
\mathcal{L}_{\text{SR}} = & - \sum_{ij} M_{ij}^{2Q} (\mathcal{Q}_i^\dagger \mathcal{Q}_j) - \sum_{ij} M_{ij}^{2\bar{U}} (\bar{\mathcal{U}}_i^\dagger \bar{\mathcal{U}}_j) - \sum_{ij} M_{ij}^{2\bar{D}} (\bar{\mathcal{D}}_i^\dagger \bar{\mathcal{D}}_j) \\
& - \sum_{ij} M_{ij}^{2L} (\mathcal{L}_i^\dagger \mathcal{L}_j) - \sum_{ij} M_{ij}^{2\bar{E}} (\bar{\mathcal{E}}_i^\dagger \bar{\mathcal{E}}_j) \\
& - (\bar{\lambda}_3 m_{\text{gluino}} \lambda_3) - (\bar{\lambda}_2 m_{\text{wino}} \lambda_2) - (\bar{\lambda}_1 m_{\text{bino}} \lambda_1) \\
& - \sum_{ij} A_{ij}^D h_{ij}^D (\mathcal{Q}_i^T e \mathcal{H}_1) \bar{\mathcal{D}}_j - \sum_{ij} A_{ij}^E h_{ij}^E (\mathcal{L}_i^T e \mathcal{H}_1) \bar{\mathcal{E}}_j \\
& - \sum_{ij} A_{ij}^U h_{ij}^U (\mathcal{Q}_i^T e \mathcal{H}_2) \bar{\mathcal{U}}_j - \sum_{ij} C_{ij}^D h_{ij}^D (\mathcal{Q}_i^T \mathcal{H}_2^*) \bar{\mathcal{D}}_j \\
& - \sum_{ij} C_{ij}^E h_{ij}^E (\mathcal{L}_i^T \mathcal{H}_2^*) \bar{\mathcal{E}}_j - \sum_{ij} C_{ij}^U h_{ij}^U (\mathcal{Q}_i^T \mathcal{H}_1^*) \bar{\mathcal{U}}_j \\
& - B\mu (\mathcal{H}_2^T e \mathcal{H}_1) + \text{H.c.}
\end{aligned} \tag{28.4.1}$$

Script letters are used here to denote the scalar components of left-chiral superfields. Sums over $SU(2)$ and color indices are understood, as necessary for invariance under $SU(3) \cdot SU(2) \cdot U(1)$, with e the usual antisymmetric $2 \cdot 2$ matrix $i\sigma_2$. All coefficients may be complex, and the gaugino masses may involve terms proportional to the γ_5 as well as the unit matrix.

We follow the custom here of writing the coefficients of the terms involving scalar fields but not their adjoints as equal to the coefficients of the corresponding supersymmetric $\mathcal{F}$ -terms in Eq. (28.1.7) times factors A_{ij}^D , A_{ij}^E , A_{ij}^U , and B . This is motivated by the consideration that the smallness of the Yukawa couplings of the light quarks in Eq. (28.1.7) reflects a number of approximate chiral symmetries that if extended to the whole supermultiplet would also make the corresponding trilinear terms in Eq. (28.4.1) small, while the appearance of the μ -term in Eq. (28.1.7) violates a possible Peccei–Quinn symmetry [4] that if approximately valid would make both μ and $B\mu$ small. Similar considerations suggest the form in which we write the coefficients of the terms that involve both scalars and their complex conjugates. Also, in Section 31.4 we will describe contributions to Ah and $B\mu$ that really are proportional to h and μ , respectively. However, here we are leaving it an open question whether the Ah , Ch , and $B\mu$ coefficients in Eq. (28.4.1) are necessarily small when the corresponding h and μ coefficients in Eq. (28.1.7) are small.

The Ch -terms in Eq. (28.4.1) have generally been omitted in discussions of the minimal supersymmetric standard model. This is partly because, as discussed in Section 27.7, terms like these that involve the ϕ -components of left-chiral scalar superfields and also their complex conjugates can potentially produce quadratic divergences and thereby raise fine-tuning problems. But we saw in Section 27.7 that the quadratic divergences occur only in ‘tadpole’ graphs in which a scalar field line disappears into the vacuum, and in the minimum supersymmetric standard model there are no scalars that are neutral under all gauge symmetries, and hence no scalar tadpoles. The Ch -terms are absent in the theories of gravitationally mediated supersymmetry breaking discussed in Section 31.6, and small

in the theories of gauge-mediated supersymmetry breaking described in Section 28.6, but there is no reason to suppose that this will always be the case.

Even though they are not supersymmetric, superrenormalizable interactions like those in Eq. (28.4.1) are shown in Section 27.7 to not produce supersymmetry-violating ultraviolet-divergent corrections to the coefficients of the supersymmetric $d = 4$ interactions. The condition of supersymmetry that is imposed on the dimensionless couplings of the minimal supersymmetric standard model therefore does not get in the way of the cancellation of ultraviolet divergences by renormalization of coupling constants. It was this property, rather than any theory of supersymmetry breaking in a high energy hidden sector, that motivated the introduction of the minimal supersymmetric standard model in Reference 10.

The best reason today for studying the implications of the supersymmetric standard model is that, as already mentioned, theories in which supersymmetry is spontaneously broken at a high energy scale are naturally described by the minimal supersymmetric standard model at much lower energies. We can explore the phenomenological implications of the supersymmetric standard model, and be reasonably confident that the results will be relevant whatever detailed model of supersymmetry breaking and its mediation turns out to be correct.

Even without the Ch -terms, if all of the other coefficients in the Lagrangian are constrained only by gauge symmetries and R parity conservation, the minimal supersymmetric standard model will contain over 100 free parameters [11]. Here ‘minimal’ means nothing more than that the theory contains only a minimal menu of superfields. Sometimes the term ‘minimal supersymmetric standard model’ is reserved for models that also satisfy restrictions on the coefficients of the superrenormalizable terms, motivated either by some underlying theory or by empirical constraints. For instance, the minimal supersymmetric standard model is sometimes optimistically assumed to satisfy the universality conditions

$$\begin{aligned} M_{ij}^{2Q} &= M_{ij}^{2\bar{D}} = M_{ij}^{2\bar{U}} = M_{ij}^{2L} = M_{ij}^{2\bar{E}} = M^2 \delta_{ij}, \\ m_{\text{gluino}} &= m_{\text{wino}} = m_{\text{bino}}, \\ A_{ij}^D &= A_{ij}^E = A_{ij}^U = A, \quad C_{ij}^D = C_{ij}^E = C_{ij}^U = 0. \end{aligned} \tag{28.4.2}$$

Often these conditions are imposed at the scale $M_X \approx 10^{16}$ GeV of coupling constant unification, with corrections produced only by the renormalization group flow to lower energies. We will not be making such assumptions here.

In analyzing the phenomenological implications of the minimal supersymmetric standard model, we must deal not only with the search for new particles, but also with two classes of severe empirical constraint on processes involving known particles: the experimental upper bounds on various flavor non-conserving processes, and on various modes of CP non-conservation.

Flavor Changing Processes

We saw in Section 21.3 that there is an automatic suppression of flavor-changing processes like $K^0\text{-}\bar{K}^0$ oscillations and $K^0 \rightarrow \mu^+\mu^-$ in the non-supersymmetric standard model. This is due to the feature of this theory, that it is only the mass splittings of the quarks

that prevent them from being defined so that each flavor is separately conserved, so the amplitude of these flavor-changing process must be proportional to several factors of small quark masses. Also, in this theory lepton flavor is automatically conserved, so that processes like $\mu \rightarrow e\gamma$ are absolutely forbidden. These satisfactory results are put at risk by the presence of squarks and sleptons in supersymmetric extensions of the standard model, because there is in general no reason to expect that the squark and slepton mass matrices will be diagonal in the same basis as the quark and lepton mass matrices. This does not introduce flavor changing in the interaction of these particles with gauge bosons, which are flavor-independent, but it can produce flavor-changing transitions in which squarks or sleptons turn into quarks or leptons with the emission or absorption of gauginos. Of course, there

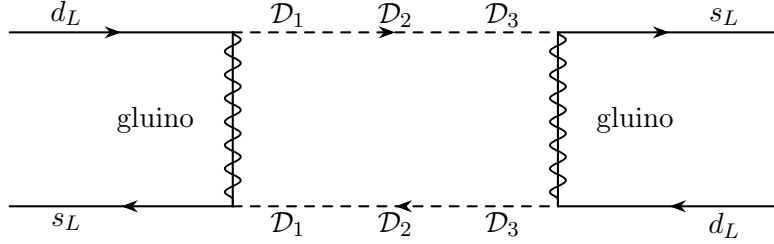


Figure 28.1. A one-loop diagram that can contribute to the $\Delta S = 2$ effective interaction $(\bar{s}_L \gamma^\mu d_L)(\bar{s}_L \gamma_\mu d_L)$ in the supersymmetric standard model. Here solid lines are quarks; dashed lines are squarks; and combined solid and wavy lines are gluinos.

is no problem if the squarks and sleptons are degenerate, in which case their mass matrices are diagonal in any basis.

The most stringent limits on squark mass splittings and/or mixing angles are set by measurements of K^0 - $\bar{K}^0$ transitions [12]. These transitions are produced by operators in the effective low-energy Lagrangian density like $(\bar{s}_L \gamma^\mu d_L)(\bar{s}_L \gamma_\mu d_L)$, which can be produced by diagrams like Figure 28.1. The superpartners of the quarks d_L and s_L are in general linear combinations $\sum_i V_{di} \mathcal{D}_i$ and $\sum_i V_{si} \mathcal{D}_i$ of squarks $\mathcal{D}_i$ of definite mass, where V_{ji} is a 3×3 unitary matrix, so the two squark propagators in this diagram contribute a factor

$$\sum_i \frac{V_{di} V_{si}^*}{k^2 + M_i^2 - i\epsilon} \cdot \sum_j \frac{V_{dj} V_{sj}^*}{k^2 + M_j^2 - i\epsilon},$$

where k is the four-momentum circulating in the loop. Because V_{ji} is unitary, this vanishes if the three squark masses M_i are all equal. If the squark square masses differ from some common value M_{squark}^2 by relatively small amounts ΔM_i^2 , then this becomes

$$\left(\frac{1}{k^2 + M_{\text{squark}}^2 - i\epsilon} \right)^4 \left(\sum_i V_{di} V_{si}^* \Delta M_i^2 \right)^2.$$

The amplitude for $d_L \bar{s}_L \rightarrow s_L \bar{d}_L$ has dimensionality mass^{-2} , so after multiplying by the gluino propagators and four factors of the strong coupling g_s , and integrating over k , we

must get an amplitude proportional to

$$\frac{g_s^4}{\widetilde{M}^6} \left(\sum_i V_{di} V_{si}^* \Delta M_i^2 \right)^2, \quad (28.4.3)$$

where $\widetilde{M}$ is the larger of M_{squark} and m_{gluino} . This may be compared with the result for this amplitude in the non-supersymmetric standard model, which is produced by W exchange, as shown in Figure 28.2. Ignoring the third generation of quarks, which has only small transition amplitudes to the first two generations, the amplitudes for $d \rightarrow u$, $d \rightarrow c$, $s \rightarrow u$ and $s \rightarrow c$ by W^- emission are, respectively, $\cos \theta_c$, $-\sin \theta_c$, $\sin \theta_c$, and $\cos \theta_c$, where θ_c is the Cabibbo angle defined in Section 21.3. Hence in place of the squark propagators here we have the quark propagators

$$\sin \theta_c \cos \theta_c \left(\frac{i\not{k} + m_u}{k^2 + m_u^2 - i\epsilon} - \frac{i\not{k} + m_c}{k^2 + m_c^2 - i\epsilon} \right),$$

and in place of the strong coupling g_s here we have the $SU(2)$ coupling g . Thus in the non-supersymmetric standard model the amplitude for $d_L \bar{s}_L \rightarrow s_L \bar{d}_L$ is proportional to

$$\frac{g^4 \sin^2 \theta_c \cos^2 \theta_c}{m_W^4} (m_c - m_u)^2, \quad (28.4.4)$$

with a proportionality coefficient of the same order as that in Eq. (28.4.3). With a plausible guess on how to calculate the K^0 - $\bar{K}^0$ transition amplitude from the amplitude for $d_L \bar{s}_L \rightarrow s_L \bar{d}_L$, the amplitude corresponding to Figure 28.2 is known to give a result in good agreement with experiment. (Indeed, Gaillard and Lee [13] used this calculation to predict that $m_c \approx 1.5$ GeV, before the c quark was discovered.) It therefore seems reasonable to require that the squark exchange result (28.4.3) should be less than the quark exchange result (28.4.4). This yields the condition

$$\left| \sum_i V_{di} V_{si}^* \frac{\Delta M_i^2}{\widetilde{M}^2} \right| < \frac{g^2}{g_s^2} \sin \theta_c \cos \theta_c \frac{(m_c - m_u) \widetilde{M}}{m_W^2}. \quad (28.4.5)$$

Taking $g^2/4\pi = 0.036$, $g_s^2/4\pi = 0.118$, $\sin \theta_c = 0.22$, $m_W = 80.4$ GeV, $m_c = 1.5$ GeV, and $m_u \ll m_c$, we find

$$\left| \sum_i V_{di} V_{si}^* \frac{\Delta M_i^2}{\widetilde{M}^2} \right| < 1.5 \cdot 10^{-3} \cdot (\widetilde{M}/100 \text{ GeV}). \quad (28.4.6)$$

The squark masses are unlikely to be much less than m_{gluino} , so we can conclude that either the squark masses are split by no more than about one part in 10^3 , or the non-diagonal terms of the mixing matrix V_{ji} are less than about 10^{-3} , or the squarks are heavier than about 10 TeV, or we have some combination of nearly degenerate squarks, nearly zero mixing angles, and heavy squarks. In itself, this result only constrains the superpartners $\mathcal{D}_i$ of the left-handed quarks of charge $-e/3$, but similar limits on the masses and mixing angles of the $\bar{\mathcal{D}}_i$ squarks can be obtained by considering the amplitude for $d_R \bar{s}_R \rightarrow s_R \bar{d}_R$. We can also obtain somewhat weaker

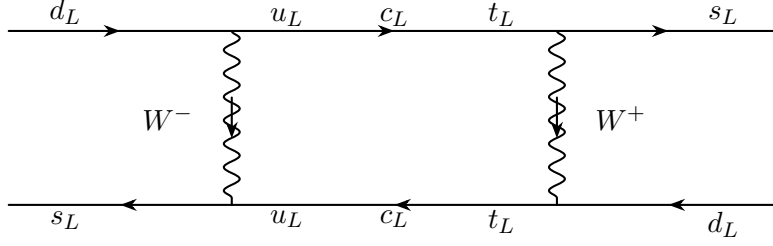


Figure 28.2. A one-loop diagram that can contribute to the $\Delta S = 2$ effective interaction $(\bar{s}_L \gamma^\mu d_L)(\bar{d}_L \gamma_\mu s_L)$ in both the supersymmetric and non-supersymmetric standard models. Here solid lines are quarks and wavy lines are $W^\pm$ bosons.

limits on the masses and mixing angles of the $\mathcal{U}_i$ squarks by considering the amplitudes produced by wino exchange rather than gluino exchange. It should be noted, however, that these arguments put no constraints on the differences between the masses of squarks of different charge, or on the differences between the masses of the superpartners $\mathcal{Q}_i$ and $\bar{\mathcal{Q}}_i$ of left-handed quarks and antiquarks.

Just as for the squarks, the sleptons of definite mass are expected to be non-diagonal linear combinations of the superpartners of the leptons. This leads to the decay process $\mu \rightarrow e + \gamma$ through diagrams like Figure 28.3. The experimental upper bound $4.9 \cdot 10^{-11}$ on the branching ratio of this process then sets a limit of about 10^{-3} on the fractional mass splitting of sleptons of the same charge but different generations for generic mixing angles, or on the mixing angles for non-degenerate sleptons [14].

Attempts have been made to explain the degeneracy of the squarks and sleptons in terms of a gauge symmetry connecting the different generations [14a]. In Section 28.6 we will describe an approach to supersymmetry breaking in which this degeneracy appears without needing to impose such symmetries.

CP violation

The second important class of constraint provided by experimental information about known particles has to do with CP-violating effects, such as the electric dipole moments of the neutron and electron [15]. In Section 21.3 we saw that these effects are rather weak in the non-supersymmetric standard model with only one scalar doublet, apart from a potential problem with the parameter θ of quantum chromodynamics, discussed in

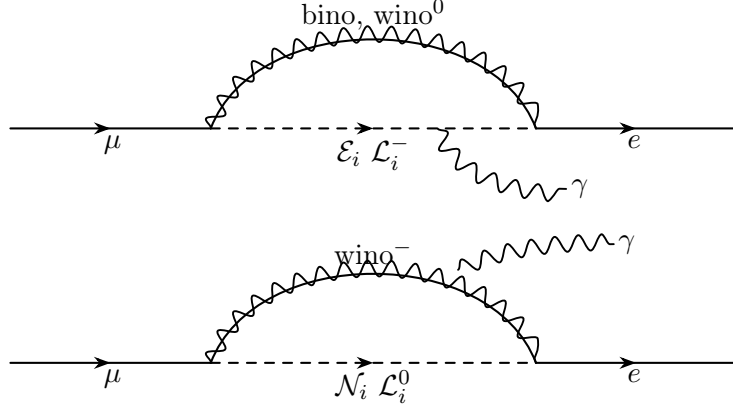


Figure 28.3. One-loop diagrams for the process $\mu \rightarrow e + \gamma$. Here solid lines are leptons; dashed lines are sleptons; combined solid and wavy lines are gauginos; and wavy lines are photons.

Section 23.6. This is because all CP-violating phases in the mass matrix of quarks and leptons and their interaction with gauge bosons could be absorbed into the definition of the quark and lepton fields if there were only two generations of quarks and leptons, and although there is a third generation, its mixing with the first two generations is (for mysterious reasons) quite weak. (This argument does not apply to processes that directly involve quarks of the third generation, such as $B^0\text{--}\bar{B}^0$ mixing, to be measured in the planned ‘B factories.’) The electric dipole moment of the neutron in this simple non-supersymmetric version of the standard model is consequently expected [16] to be less than about $10^{-30} e \text{ cm}$, well below the experimental upper bound, $6.3 \cdot 10^{-26} e \text{ cm}$ [16a].

In contrast, the over 100 parameters of the minimum supersymmetric standard model in its most general form involve dozens of CP-violating relative phases. After integrating out the heavy superpartners of the known particles, these phases produce a number of CP-violating effective interactions to be added to the Lagrangian of the standard model. Those of minimum dimensionality, which on dimensional grounds are likely to be the most important, include electric dipole moments of the quarks and leptons [17], similar CP-violating ‘chromoelectric’ dipole moments

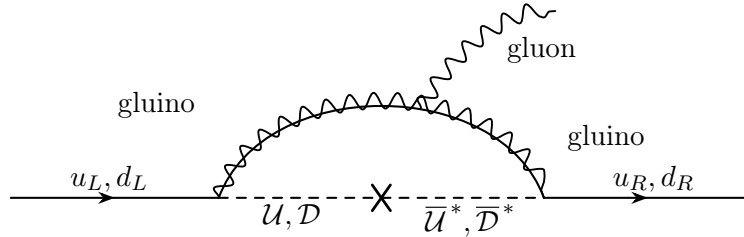


Figure 28.4. A one-loop diagram for the chromoelectric dipole moment of the u or d quarks. Here solid lines are quarks; dashed lines are squarks; combined solid and wavy lines are gluinos; and the wavy line is a gluon. The X represents the insertion of a bilinear interaction arising from a trilinear scalar field interaction combined with the spontaneous breakdown of $SU(2) \cdot U(1)$. There are also diagrams in which the gluon line is attached to one of the internal squark lines instead of to the gluino line.

contributing to the interactions of gluons with quarks [18], a CP-violating purely gluonic interaction [19], and a CP-violating interaction of the lightest Higgs scalar with leptons [20].

To take one example, consider the quark chromoelectric dipole moments, which in some models make the largest contribution to the electric dipole moment of the neutron. The CP-violating chromoelectric dipole moment operator is $(\bar{q}\gamma_5[\gamma_\mu, \gamma_\nu]\lambda_a q)f_a^{\mu\nu}$ (where q is a u or d color-triplet quark field, $f_a^{\mu\nu}$ is the $SU(3)$ field-strength tensor, and λ_a are the $3 \cdot 3$ generators of $SU(3)$). Since $\gamma_5[\gamma_\mu, \gamma_\nu]$ has matrix elements only between $\bar{q}_L$ and q_R or between $\bar{q}_R$ and q_L , in order for a one-loop graphs to contribute to the chromoelectric dipole moment an external left-handed u or d quark line must emit an internal gluino line and turn into a $\mathcal{U}$ or $\mathcal{D}$ squark line, which next turns into a $\bar{\mathcal{U}}^*$ or $\bar{\mathcal{D}}^*$ squark line, and then into a right-handed u or d quark line by absorbing the internal gluino line, with the external gluon line attached either to the internal gluino line or to one of the internal squark lines. (See Figure 28.4.)

To calculate this, we need to know the mixing of the scalar components $\mathcal{U}_i$ (or $\mathcal{D}_i$) of the left-chiral quark superfields Q_i with the complex conjugates $\bar{\mathcal{U}}_j^*$ (or $\bar{\mathcal{D}}_j^*$) of the scalar components of the left-chiral antiquark superfields $\bar{Q}_j$ (or $\bar{D}_j$), produced by the spontaneous breakdown of $SU(2) \cdot U(1)$, and represented by the X in Figure 28.4. Part of this mixing arises from a contribution of the supersymmetric $\mathcal{F}$ -term interaction (28.1.7) to the last term in Eq. (26.4.7):

$$\mathcal{L}_{\mathcal{Q}\bar{\mathcal{Q}}\mathcal{H}} = - \left| \sum_{ij} h_{ij}^U \mathcal{U}_i \bar{\mathcal{U}}_j + \mu \mathcal{H}_1^0 \right|^2 - \left| \sum_{ij} h_{ij}^D \mathcal{D}_i \bar{\mathcal{D}}_j + \mu \mathcal{H}_2^0 \right|^2. \quad (28.4.7)$$

There is also a contribution from the A - and C -terms in Eq. (28.4.1):

$$\begin{aligned} \mathcal{L}'_{\mathcal{Q}\bar{\mathcal{Q}}\mathcal{H}} = & - \sum_{ij} h_{ij}^D \mathcal{D}_i \bar{\mathcal{D}}_j [-A_{ij}^D \mathcal{H}_1^0 + C_{ij}^D \mathcal{H}_2^{0*}] \\ & - \sum_{ij} h_{ij}^U \mathcal{U}_i \bar{\mathcal{U}}_j [A_{ij}^U \mathcal{H}_2^0 + C_{ij}^U \mathcal{H}_1^{0*}] - \text{H.c.} \end{aligned} \quad (28.4.8)$$

Replacing the neutral Higgs scalar fields with their expectation values gives the quadratic terms

$$\begin{aligned} \mathcal{L}_{\mathcal{Q}\bar{\mathcal{Q}}} = & -2 \text{Re} \sum_{ij} m_{ij}^U \mathcal{U}_i \bar{\mathcal{U}}_j (\mu^* \cot \beta + A_{ij}^U + C_{ij}^U \cot \beta) \\ & -2 \text{Re} \sum_{ij} m_{ij}^D \mathcal{D}_i \bar{\mathcal{D}}_j (\mu^* (\tan \beta)^* + A_{ij}^D - C_{ij}^D (\tan \beta)^*), \end{aligned} \quad (28.4.9)$$

where $m_{ij}^U = \langle \mathcal{H}_2^0 \rangle h_{ij}^U$ and $m_{ij}^D = -\langle \mathcal{H}_1^0 \rangle h_{ij}^D$ are the mass matrices of the quarks of charge $2e/3$ and $-e/3$, and

$$\tan \beta \equiv \frac{\langle \mathcal{H}_2^0 \rangle}{\langle \mathcal{H}_1^0 \rangle^*}. \quad (28.4.10)$$

Neglecting Cabibbo mixing, and for definiteness taking the A s and C s diagonal, Figure 28.4 makes contributions to the chromoelectric dipole moment of the u and d quarks of the form

$$d_u^{ce} = \frac{g_s^3}{16\pi^2} \text{Im} [m_u A'_u I(m_{\mathcal{U}}, m_{\bar{\mathcal{U}}}, m_{\text{gluino}})], \quad (28.4.11)$$

$$d_d^{ce} = \frac{g_s^3}{16\pi^2} \text{Im} [m_d A'_d I(m_{\mathcal{D}}, m_{\overline{\mathcal{D}}}, m_{\text{gluino}})], \quad (28.4.12)$$

where

$$A'_u \equiv (\mu^* + C_u) \cot \beta + A_u, \quad A'_d \equiv (\mu^* - C_d)(\tan \beta)^* + A_d, \quad (28.4.13)$$

and I is a complicated dimensionless function of its arguments arising from the integration over the virtual four-momentum. For $m_{\mathcal{Q}} \simeq m_{\overline{\mathcal{Q}}}$ and the gluino field defined to make m_{gluino} as well as $m_{\mathcal{Q}}$ real, the function I takes the form

$$I(m_{\mathcal{Q}}, m_{\mathcal{Q}}, m_{\text{gluino}}) = m_{\text{gluino}}^{-3} J \left(\frac{m_{\text{gluino}}^2}{m_{\mathcal{Q}}^2 - m_{\text{gluino}}^2} \right), \quad (28.4.14)$$

where [21]

$$J(z) = 2 \left(-z^4 + \frac{4}{3}z^3 + z^2 \right) \ln \left(\frac{1+z}{z} \right) + 2z^3 - \frac{11}{3}z^2. \quad (28.4.15)$$

The hard part of this sort of calculation always lies in estimating the contribution of an operator like the chromoelectric dipole interaction to hadronic matrix elements such as the electric dipole moment of the neutron. We can anticipate that there will be renormalization group corrections required because this operator is to be used at energies of the order of the neutron mass, rather than the masses of the squarks and gluino. More important is simply getting dimensional factors and factors of 4π right. For this purpose, it is usual to use a counting rule [22] known as ‘naive dimensional analysis.’ A connected graph with V_i vertices of type i and I internal lines will have a number L of loops given by $L = I - \sum_i V_i + 1$. If there are N_i lines attached to a vertex of type i and N external lines of the whole graph, then $2I + N = \sum_i V_i N_i$, so

$$L = 1 - \frac{N}{2} + \sum_i V_i \left(\frac{N_i}{2} - 1 \right).$$

We expect a factor of order $1/16\pi^2$ for each loop, so the coefficient of an operator $\mathcal{O}$ in the low-energy effective Lagrangian with N field factors will contain an over-all factor

$$(4\pi)^{N-2} \prod_i (4\pi)^{(2-N_i)V_i}.$$

If the operator $\mathcal{O}$ has dimensionality d and the interactions $\mathcal{O}_i$ of type i have dimensionality d_i , then the coefficient of $\mathcal{O}$ will have dimensionality $4 - d - \sum_i (4 - d_i)$, so this coefficient will also have a factor $M^{4-d} \prod_i M^{d_i-4}$, where M is some scale that is typical of hadronic physics, such as the nucleon mass or the energy $2\pi F_\pi \simeq 1200$ MeV, where the low-energy expansions discussed in Section 19.5 begin to break down. Finally, the contribution to the coefficient of $\mathcal{O}$ from some graph will of course be proportional to the couplings of all the operators $\mathcal{O}_i$ associated with the vertices in the graph. These remarks can be conveniently summarized by defining a ‘reduced coupling’: the reduced coupling associated with any operator $\mathcal{O}_i$ having N_i field factors, dimensionality d_i , and coupling constant g_i is

$$g_i^{\text{reduced}} \equiv g_i (4\pi)^{2-N_i} M^{d_i-4}. \quad (28.4.16)$$

The above estimates suggest the rule of naive dimensional analysis: the reduced coupling of any operator $\mathcal{O}$ in the effective hadronic Lagrangian is roughly equal to the product of the reduced couplings of the interactions that contribute to this effective coupling.

The neutron electric dipole moment is the coefficient of an operator with one photon and two neutron fields and dimensionality 5, so its reduced coupling is $Md_n^e/4\pi$. Likewise, the quark chromoelectric magnetic moment has reduced coupling $Md_q^{ce}/4\pi$. In addition to one factor of this reduced coupling, the reduced coupling of the neutron electric dipole operator must have a factor of the reduced coupling $e/4\pi$ of the electromagnetic coupling and an indeterminate number of factors of the reduced strong coupling $g_s/4\pi$, which at the low-energy scale M are not very different from unity and will be ignored. Taking the d quark contribution as representative of the contributions of both u and d quarks, the result then is that

$$d_n^e \approx \frac{ed_d^{ce}}{4\pi} \approx e \left(\frac{g_s}{4\pi} \right)^3 \text{Im} [m_d A_d'] I(m_{\mathcal{D}}, m_{\overline{\mathcal{D}}}, m_{\text{gluino}}). \quad (28.4.17)$$

Further simplifying by setting $m_{\text{gluino}} \simeq m_{\mathcal{D}} \simeq m_{\overline{\mathcal{D}}}$, so that $J = 7/18$, and taking $g_s^2/4\pi$ at the scale of squark and gluino masses to have the same value 0.12 as at m_Z , and $|m_d| \approx 7$ MeV, we have then

$$|d_n^e| \approx 0.5 \cdot 10^{-23} e \text{ cm} \frac{|A_d'| |\sin \varphi| \cdot (100 \text{ GeV})^2}{m_{\text{gluino}}^3}, \quad (28.4.18)$$

where φ is the phase of A_d' , with the convention that the gluino, quark, and squark masses are taken real. The contribution of the electric dipole moment of the quark is somewhat larger, while the contribution of the purely gluonic CP-odd operator is considerably smaller [23].

To avoid conflict with the experimental upper bound of $0.97 \cdot 10^{-25} e \text{ cm}$, either the phases associated with CP violation in the supersymmetric standard model must be less than about 10^{-2} , or some of the new particles of this model must be heavier than about 1 TeV. Similar conclusions have been reached from calculations of the electric dipole moments of atoms and molecules [23]. Even more stringent conditions on CP-violating phases have been derived [24] by considering the contribution of Figure 28.1 to the one precisely measured CP-violating effect, the imaginary part of the amplitude for K^0 - $\bar{K}^0$ oscillation.

28.5 The Sector of Zero Baryon and Lepton Number

Despite the large number of parameters of the supersymmetric standard model, in some contexts it is surprisingly predictive. This is true in particular when we consider the scalar fields whose vacuum expectation values spontaneously break the $SU(2) \cdot U(1)$ gauge symmetry. In this section we will consider these scalars, along with other fields of zero baryon and lepton number: neutral scalars odd under charge conjugation, charged scalars, and the fermionic superpartners of these scalars and of the $W^\pm$ and Z^0 .

It is a crucial requirement for supersymmetric versions of the standard model that they should contain scalar doublet ‘Higgs’ superfields with the right mass and interaction parameters to account for the breakdown of the $SU(2) \cdot U(1)$ gauge group of the electromagnetic and weak interactions. We saw in Section 28.1 that at least two left-chiral scalar doublets

are needed to give mass to the quarks of both charge $2e/3$ and $-e/3$ and the charged leptons, while we found in Section 28.2 that two doublets are just what is needed to bring the $SU(3)$, $SU(2)$, and $U(1)$ gauge couplings together at some very high energy. We have therefore assumed that there are two left-chiral scalar $SU(2)$ doublets

$$H_1 = \begin{pmatrix} H_1^0 \\ H_1^- \end{pmatrix}, \quad H_2 = \begin{pmatrix} H_2^+ \\ H_2^0 \end{pmatrix}. \quad (28.5.1)$$

These have $SU(2)$ and $U(1)$ D -terms (27.4.7) given (assuming zero Fayet–Iliopoulos constant $\xi_{U(1)}$) by

$$D = \frac{g}{2} (\mathcal{H}_1^\dagger \tau \mathcal{H}_1) + \frac{g}{2} (\mathcal{H}_2^\dagger \tau \mathcal{H}_2), \quad (28.5.2)$$

$$D_y = \frac{g'}{2} (\mathcal{H}_1^\dagger \mathcal{H}_1) - \frac{g'}{2} (\mathcal{H}_2^\dagger \mathcal{H}_2), \quad (28.5.3)$$

where $\mathcal{H}_{1,2}$ are the scalar components of the superfield doublets $H_{1,2}$ and τ_r are the Pauli matrices, with $\tau_r^2 = \mathbf{1}$. As shown in Eq. (27.4.9), in renormalizable theories this gives the D -term contribution to the scalar field potential

$$\begin{aligned} V_D &= \frac{1}{2} D^2 + \frac{1}{2} D_y^2 \\ &= \frac{g^2}{8} \left[(\mathcal{H}_1^\dagger \tau \mathcal{H}_1) + (\mathcal{H}_2^\dagger \tau \mathcal{H}_2) \right]^2 + \frac{g'^2}{8} \left[(\mathcal{H}_1^\dagger \mathcal{H}_1) - (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]^2. \end{aligned} \quad (28.5.4)$$

This can be put in a more convenient form by using the relation

$$(\tau)_{i\ell} \cdot (\tau)_{kj} = 2\delta_{ij}\delta_{k\ell} - \delta_{i\ell}\delta_{kj}. \quad (28.5.5)$$

(To prove this, use rotational invariance to show that $\delta_{ij}\delta_{k\ell}$ may be expressed as a linear combination of $(\tau)_{i\ell} \cdot (\tau)_{kj}$ and $\delta_{i\ell}\delta_{kj}$, and calculate the coefficients by taking traces on the indices i, j and i, ℓ .) In this way, we may rewrite the D -term part of the scalar field potential as

$$V_D = \frac{g^2}{2} |\mathcal{H}_1^\dagger \mathcal{H}_2|^2 + \frac{g^2 + g'^2}{8} \left[(\mathcal{H}_1^\dagger \mathcal{H}_1) - (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]^2. \quad (28.5.6)$$

As mentioned in Section 28.1, there is just one possible renormalizable term in the superpotential for these two left-chiral doublets, of the form

$$f(H_1, H_2) = \mu (H_1^T e H_2), \quad (28.5.7)$$

where μ is a constant with the dimensions of a mass and e is the antisymmetric matrix $i\tau_2$. According to Eq. (27.4.9), this gives an additional contribution to the scalar field potential

$$\begin{aligned} V_\mu &= \sum_r \left| \frac{\partial f(\mathcal{H}_1, \mathcal{H}_2)}{\partial \mathcal{H}_{1r}} \right|^2 + \sum_r \left| \frac{\partial f(\mathcal{H}_1, \mathcal{H}_2)}{\partial \mathcal{H}_{2r}} \right|^2 \\ &= |\mu|^2 \left[(\mathcal{H}_1^\dagger \mathcal{H}_1) + (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]. \end{aligned} \quad (28.5.8)$$

For $\mu \neq 0$ the potential $V_D + V_\mu$ evidently has a minimum value of zero, reached at the unique point $\mathcal{H}_1 = \mathcal{H}_2 = 0$. With just these terms in the potential, $SU(2) \cdot U(1)$ as well as

supersymmetry is not spontaneously broken. (The case $\mu = 0$ is not much better; there is a continuous infinity of vacuum states with supersymmetry unbroken and with $SU(2) \cdot U(1)$ broken down to electromagnetic gauge invariance with all possible strengths, including zero.) This is one more example of the general difficulty, already seen in Section 28.3, of formulating realistic theories in which supersymmetry is spontaneously broken within the standard model.

Under the assumption of the previous section that supersymmetry is violated in the effective Lagrangian only by superrenormalizable terms, the most general such supersymmetry-breaking term involving the scalar doublets is of the form

$$V_m = m_1^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + m_2^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) + \text{Re} \{ B\mu (\mathcal{H}_1^T e \mathcal{H}_2) \},$$

where m_1^2 and m_2^2 are real parameters (not necessarily positive) and $B\mu$ is a parameter of arbitrary phase. We will adjust the over-all phase of the superfields H_1 and H_2 so that $B\mu$ is real and positive, and so

$$V_m = m_1^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + m_2^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) + B\mu \text{Re} (\mathcal{H}_1^T e \mathcal{H}_2). \quad (28.5.9)$$

The total scalar potential in the tree approximation is then

$$\begin{aligned} V &= V_D + V_\mu + V_m \\ &= \frac{g^2}{2} |\mathcal{H}_1^\dagger \mathcal{H}_2|^2 + \frac{g^2 + g'^2}{8} \left[(\mathcal{H}_1^\dagger \mathcal{H}_1) - (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]^2 \\ &\quad + (m_1^2 + |\mu|^2) (\mathcal{H}_1^\dagger \mathcal{H}_1) + (m_2^2 + |\mu|^2) (\mathcal{H}_2^\dagger \mathcal{H}_2) \\ &\quad + B\mu \text{Re} (\mathcal{H}_1^T e \mathcal{H}_2). \end{aligned} \quad (28.5.10)$$

Note in particular that μ^2 , m_1^2 , and m_2^2 only appear in the combinations $m_1^2 + |\mu|^2$ and $m_2^2 + |\mu|^2$.

There is one condition on the supersymmetry-breaking parameters m_i^2 , derived from the requirement that the potential should be bounded below. For scalar fields going to infinity in generic directions, the potential is dominated by the quartic terms V_D , which are positive. There are special directions in which V_D vanishes: those for which (up to an $SU(2) \cdot U(1)$ gauge transformation)

$$\mathcal{H}_1 = \begin{pmatrix} \phi \\ 0 \end{pmatrix}, \quad \mathcal{H}_2 = \begin{pmatrix} 0 \\ \phi \end{pmatrix},$$

with ϕ an arbitrary complex quantity. For such directions, $V = (2|\mu|^2 + m_1^2 + m_2^2)|\phi|^2 - B\mu\phi^2$, so (since $B\mu$ has been defined to be positive) in order for this not to go to $-\infty$ as $\phi \rightarrow +\infty$, it is necessary that

$$2|\mu|^2 + m_1^2 + m_2^2 \geq B\mu. \quad (28.5.11)$$

We wish to look for a minimum of the potential at which electromagnetic gauge invariance is not broken, so let us consider the behavior of the potential as a function of the neutral

scalar fields, with the charged scalar fields set equal to zero. In this case Eq. (28.5.10) gives the potential of the neutral scalars as

$$V^N = \frac{g^2 + g'^2}{8} [|\mathcal{H}_1^0|^2 - |\mathcal{H}_2^0|^2]^2 + (m_1^2 + |\mu|^2)|\mathcal{H}_1^0|^2 + (m_2^2 + |\mu|^2)|\mathcal{H}_2^0|^2 - B\mu \operatorname{Re}(\mathcal{H}_1^0 \mathcal{H}_2^0). \quad (28.5.12)$$

To find a stationary point, we expand V^N around the constant values $\mathcal{H}_i^0 = v_i$, writing

$$\mathcal{H}_i^0 = v_i + \varphi_i. \quad (28.5.13)$$

To second order in the φ_i , Eq. (28.5.12) gives

$$\begin{aligned} V_{\text{quad}}^N = & \frac{g^2 + g'^2}{4} (|v_1|^2 - |v_2|^2) [2 \operatorname{Re}(v_1^* \varphi_1 - v_2^* \varphi_2) + |\varphi_1|^2 - |\varphi_2|^2] \\ & + \frac{g^2 + g'^2}{2} [\operatorname{Re}(v_1^* \varphi_1 - v_2^* \varphi_2)]^2 \\ & + (m_1^2 + |\mu|^2) (2 \operatorname{Re} v_1^* \varphi_1 + |\varphi_1|^2) \\ & + (m_2^2 + |\mu|^2) (2 \operatorname{Re} v_2^* \varphi_2 + |\varphi_2|^2) \\ & - B\mu \operatorname{Re}(v_1 \varphi_2 + v_2 \varphi_1 + \varphi_1 \varphi_2) + \text{constant}. \end{aligned} \quad (28.5.14)$$

For the v_i to be equilibrium values of the fields, the terms of first order in the φ_i must vanish

$$(m_1^2 + |\mu|^2)v_1^* + \frac{g^2 + g'^2}{4} (|v_1|^2 - |v_2|^2)v_1^* - \frac{1}{2}B\mu v_2 = 0, \quad (28.5.15)$$

$$(m_2^2 + |\mu|^2)v_2^* + \frac{g^2 + g'^2}{4} (|v_2|^2 - |v_1|^2)v_2^* - \frac{1}{2}B\mu v_1 = 0. \quad (28.5.16)$$

Without changing the over-all phase of the φ_i , we may adjust their relative phases so that v_1 is real. Then Eqs. (28.5.15) and (28.5.16) show that v_2 is also real, so that these equations become

$$(m_1^2 + |\mu|^2)v_1 + \frac{g^2 + g'^2}{4} (v_1^2 - v_2^2)v_1 - \frac{1}{2}B\mu v_2 = 0, \quad (28.5.17)$$

$$(m_2^2 + |\mu|^2)v_2 + \frac{g^2 + g'^2}{4} (v_2^2 - v_1^2)v_2 - \frac{1}{2}B\mu v_1 = 0. \quad (28.5.18)$$

These conditions may be used to express the mass parameters in the potential in terms of the convenient quantities

$$\tan \beta \equiv v_2/v_1, \quad (28.5.19)$$

$$m_Z^2 \equiv \frac{1}{2}(g^2 + g'^2)(v_1^2 + v_2^2), \quad (28.5.20)$$

and

$$m_A^2 \equiv 2|\mu|^2 + m_1^2 + m_2^2. \quad (28.5.21)$$

(The parameter m_Z is the mass of the Z vector boson.³ We will soon see that m_A is the mass of one of the physical scalars.) Multiplying Eqs. (28.5.17) and (28.5.18), respectively, by v_2 and v_1 and taking the sum and difference gives

$$B\mu = m_A^2 \sin 2\beta, \quad (28.5.22)$$

³There is a difference of a factor of 2 between the formulas for m_Z^2 given by Eqs. (28.5.20) and (21.3.30), due to the fact that the scalar fields are normalized differently here and in Section 21.3.

and

$$m_1^2 - m_2^2 = -(m_A^2 + m_Z^2) \cos 2\beta, \quad (28.5.23)$$

which with Eq. (28.5.21) yields

$$\begin{aligned} m_1^2 + |\mu|^2 &= \frac{1}{2}m_A^2 - \frac{1}{2}(m_A^2 + m_Z^2) \cos 2\beta, \\ m_2^2 + |\mu|^2 &= \frac{1}{2}m_A^2 + \frac{1}{2}(m_A^2 + m_Z^2) \cos 2\beta. \end{aligned} \quad (28.5.24)$$

With linear terms cancelling, the quadratic part (28.5.14) of the neutral scalar potential may then be written

$$\begin{aligned} V_{\text{quad}}^N &= \frac{g^2 + g'^2}{4} (v_1^2 - v_2^2) [|\varphi_1|^2 - |\varphi_2|^2] + \frac{g^2 + g'^2}{2} [\text{Re}(v_1 \varphi_1 - v_2 \varphi_2)]^2 \\ &\quad + (m_1^2 + |\mu|^2) |\varphi_1|^2 + (m_2^2 + |\mu|^2) |\varphi_2|^2 - B\mu \text{Re}(\varphi_1 \varphi_2) + \text{constant} \\ &= \frac{1}{2}m_Z^2 \cos 2\beta [|\varphi_1|^2 - |\varphi_2|^2] + m_Z^2 [\text{Re}(\cos \beta \varphi_1 - \sin \beta \varphi_2)]^2 \\ &\quad + \frac{1}{2}m_A^2 (|\varphi_1|^2 + |\varphi_2|^2) - \frac{1}{2}(m_A^2 + m_Z^2) \cos 2\beta [|\varphi_1|^2 - |\varphi_2|^2] \\ &\quad - m_A^2 \sin 2\beta \text{Re}(\varphi_1 \varphi_2) + \text{constant}. \end{aligned} \quad (28.5.25)$$

We see from Eq. (28.5.25) that the real and imaginary parts of the φ_i are decoupled. (This is because the potential (28.5.12) is invariant under a charge conjugation or CP transformation $\varphi_i \rightarrow \varphi_i^*$.) The mass-squared matrix of the imaginary parts of the φ_i is

$$M_{\text{Im } \varphi}^2 = \begin{pmatrix} \frac{1}{2}m_A^2(1 - \cos 2\beta) & \frac{1}{2}m_A^2 \sin 2\beta \\ \frac{1}{2}m_A^2 \sin 2\beta & \frac{1}{2}m_A^2(1 + \cos 2\beta) \end{pmatrix}. \quad (28.5.26)$$

The determinant vanishes, so one eigenvalue is zero, and the other equals the trace, which is just m_A^2 . The zero-mass scalar is of course the neutral Goldstone boson associated with the spontaneous breakdown of $SU(2) \cdot U(1)$ to electromagnetic gauge invariance, and as discussed in Chapter 21 it is eliminated by the Higgs mechanism. As promised, m_A is the mass of one of the physical scalars, the non-Goldstone boson with C negative. This shows that for the field value $\varphi_i = v_i$ to be at least a local minimum of the potential, the parameter m_A^2 defined by Eq. (28.5.21) must be positive. The condition (28.5.11) for good behavior at large field strengths shows that Eq. (28.5.22) has a solution here for β in the range $0 \leq \beta \leq \pi/2$.

In particular, if $B\mu = 0$ and $0 < \beta < \pi/2$, then Eq. (28.5.22) shows that $m_A = 0$. In this case, the particle A is the Goldstone boson of a $U(1)$ Peccei–Quinn symmetry [4] of the potential (28.5.12) under an equal phase change of $\mathcal{H}_1^0$ and $\mathcal{H}_2^0$, which for $v_1 \neq 0$ and $v_2 \neq 0$ is spontaneously broken without leaving any combination of this and the electroweak $U(1)$ symmetry unbroken. This is the original version of the axion [25], which as we saw in Section 23.6 acquires only a small mass from the Yukawa interactions of the scalars with the quarks and is experimentally ruled out. Thus we can conclude that $B\mu$ definitely does not vanish.

The elements of the mass-squared matrix for the real scalars are given by Eq. (28.5.25) as

$$\begin{aligned}(M_{\text{Re}\varphi}^2)_{11} &= \frac{1}{2}m_A^2(1 - \cos 2\beta) + \frac{1}{2}m_Z^2(1 + \cos 2\beta), \\(M_{\text{Re}\varphi}^2)_{12} &= (M_{\text{Re}\varphi}^2)_{21} = -\frac{1}{2}(m_A^2 + m_Z^2)\sin 2\beta, \\(M_{\text{Re}\varphi}^2)_{22} &= \frac{1}{2}m_A^2(1 + \cos 2\beta) + \frac{1}{2}m_Z^2(1 - \cos 2\beta).\end{aligned}\tag{28.5.27}$$

Solving the secular equation gives the eigenvalues

$$m_H^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 + \sqrt{(m_A^2 + m_Z^2)^2 - 4m_A^2 m_Z^2 \cos^2 2\beta} \right], \tag{28.5.28}$$

$$m_h^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 - \sqrt{(m_A^2 + m_Z^2)^2 - 4m_A^2 m_Z^2 \cos^2 2\beta} \right]. \tag{28.5.29}$$

To calculate the masses of the charged scalars, we evaluate the potential V with the neutral scalars set equal to their vacuum expectation values:

$$\mathcal{H}_1 = \begin{pmatrix} v_1 \\ \mathcal{H}_1^- \end{pmatrix}, \quad \mathcal{H}_2 = \begin{pmatrix} \mathcal{H}_2^+ \\ v_2 \end{pmatrix}. \tag{28.5.30}$$

Using this in Eq. (28.5.10) then gives the quadratic part of the charged scalar potential

$$\begin{aligned}V_{\text{quad}}^{\text{C}} &= \frac{g^2}{2} |v_2(\mathcal{H}_1^-)^* + v_1\mathcal{H}_2^+|^2 + \frac{g^2 + g'^2}{4}(v_1^2 - v_2^2)(|\mathcal{H}_1^-|^2 - |\mathcal{H}_2^+|^2) \\&\quad + (m_1^2 + |\mu|^2)|\mathcal{H}_1^-|^2 + (m_2^2 + |\mu|^2)|\mathcal{H}_2^+|^2 + B\mu\mathcal{H}_1^-\mathcal{H}_2^+.\end{aligned}\tag{28.5.31}$$

Using Eqs. (28.5.22) and (28.5.24), this may be written

$$\begin{aligned}V_{\text{quad}}^{\text{C}} &= \frac{1}{2}(m_W^2 + m_A^2)[|\mathcal{H}_1^-|^2(1 - \cos 2\beta) + |\mathcal{H}_2^+|^2(1 + \cos 2\beta) \\&\quad + 2\sin 2\beta \mathcal{H}_1^-\mathcal{H}_2^+],\end{aligned}\tag{28.5.32}$$

where m_W is the charged gauge boson mass:

$$m_W^2 = \frac{1}{2}g^2(|v_1|^2 + |v_2|^2). \tag{28.5.33}$$

The charged scalar mass matrix is then

$$M_C^2 = \frac{1}{2}(m_W^2 + m_A^2) \begin{pmatrix} 1 - \cos 2\beta & \sin 2\beta \\ \sin 2\beta & 1 + \cos 2\beta \end{pmatrix}. \tag{28.5.34}$$

This has determinant zero, so it has one eigenvalue equal to zero and the other eigenvalue equal to the trace

$$m_C^2 = m_W^2 + m_A^2. \tag{28.5.35}$$

The zero-mass charged scalar is of course the other Goldstone boson associated with the spontaneous breakdown of $SU(2) \cdot U(1)$, and like the neutral Goldstone boson found earlier it is eliminated by the Higgs mechanism.

Even without knowing the parameters m_A and β , these results tell us a lot about the relative magnitudes of the scalar boson masses. We can rewrite Eqs. (28.5.28) and (28.5.29) in the form

$$m_H^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 + \sqrt{(m_A^2 - m_Z^2)^2 + 4m_A^2 m_Z^2 \sin^2 2\beta} \right], \quad (28.5.36)$$

$$m_h^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 - \sqrt{(m_A^2 - m_Z^2)^2 + 4m_A^2 m_Z^2 \sin^2 2\beta} \right]. \quad (28.5.37)$$

We see that *the heavier neutral scalar mass m_H is larger than the larger of m_Z and m_A , while the lighter neutral scalar mass m_h is smaller than the smaller of m_Z and m_A* . If the large ratio of top to bottom quark masses is due to a large ratio $v_2/v_1 = \tan \beta$ of scalar field vacuum expectation values, rather than a large ratio of Yukawa couplings, then we expect β to be near $\pi/2$, in which case these inequalities become approximate equalities. Further, Eq. (28.5.35) shows that *the charged scalar mass is greater than both m_A and m_W* .

These results are quantitatively modified by various radiative corrections *within* the standard model (as opposed to radiative effects that produce the input parameters m_i^2 in theories of gauge-mediated supersymmetry breaking). The most important corrections arise from the presence in the scalar potential V of terms arising from graphs consisting of a single loop of top or bottom quarks interacting any number of times with external scalar field lines. This is because the top and bottom quarks have by far the strongest couplings to $\mathcal{H}_2$ and $\mathcal{H}_1$, respectively. (It is prudent to include bottom as well as top quark loops here because, as mentioned above, the larger mass of the top quark may be due to a large ratio v_2/v_1 , rather than to a large ratio of Yukawa couplings, but even in this case we will see that the dominant corrections are due to top quark loops.)

Let us first consider the neutral scalars, at least one of which would be lighter than the Z boson in the absence of radiative corrections. The effect of these top and bottom loops is to contribute a term of form $U_t(|\mathcal{H}_2^0|^2) + U_b(|\mathcal{H}_1^0|^2)$ to V^N . We will absorb any terms in U_b or U_t that are linear in $|\mathcal{H}_1^0|^2 - v_1^2$ or $|\mathcal{H}_2^0|^2 - v_2^2$ into the input parameters m_1^2 and m_2^2 , so that

$$U'_b(v_1^2) = U'_t(v_2^2) = 0. \quad (28.5.38)$$

Our earlier results (28.5.24) and (28.5.22) for $m_1^2 + |\mu|^2$, $m_2^2 + |\mu|^2$ and $B\mu$ are then unchanged. Also, the mass matrix for the C -odd neutral scalars is still given by Eq. (28.5.26). On the other hand, the elements of the mass-squared matrix for the C -even neutral scalars are now given by

$$\begin{aligned} (M_{\text{Re}\varphi}^2)_{11} &= \frac{1}{2}m_A^2(1 - \cos 2\beta) + \frac{1}{2}m_Z^2(1 + \cos 2\beta) + \Delta_b, \\ (M_{\text{Re}\varphi}^2)_{12} &= (M_{\text{Re}\varphi}^2)_{21} = -\frac{1}{2}(m_A^2 + m_Z^2) \sin 2\beta, \\ (M_{\text{Re}\varphi}^2)_{22} &= \frac{1}{2}m_A^2(1 + \cos 2\beta) + \frac{1}{2}m_Z^2(1 - \cos 2\beta) + \Delta_t, \end{aligned} \quad (28.5.39)$$

where

$$\Delta_b = 2v_1^2 U_b''(v_1^2), \quad \Delta_t = 2v_2^2 U_t''(v_2^2). \quad (28.5.40)$$

The solutions of the secular equation are then

$$m_H^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 + \Delta_t + \Delta_b + \sqrt{((m_A^2 - m_Z^2) \cos 2\beta + \Delta_t - \Delta_b)^2 + (m_A^2 + m_Z^2)^2 \sin^2 2\beta} \right], \quad (28.5.41)$$

$$m_h^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 + \Delta_t + \Delta_b - \sqrt{((m_A^2 - m_Z^2) \cos 2\beta + \Delta_t - \Delta_b)^2 + (m_A^2 + m_Z^2)^2 \sin^2 2\beta} \right]. \quad (28.5.42)$$

In considering searches for these particles, it is important to note that the light Higgs mass m_h increases as the unknown mass m_A increases, reaching a finite upper bound for $m_A \rightarrow \infty$

$$m_h \leq m_h(m_A \rightarrow \infty) = m_Z^2 \cos^2 2\beta + \Delta_t \sin^2 \beta + \Delta_b \cos^2 \beta. \quad (28.5.43)$$

To calculate Δ_b and Δ_t , we recall from Section 16.2 that the potentials U_b and U_t are given by

$$U_b(|\mathcal{H}_1^0|^2) = -\frac{3}{16\pi^2} |\lambda_b \mathcal{H}_1^0|^4 \left[\ln \frac{|\lambda_b \mathcal{H}_1^0|^2}{M_{sb}^2} - \frac{3}{2} \right] + \text{linear terms}, \quad (28.5.44)$$

$$U_t(|\mathcal{H}_2^0|^2) = -\frac{3}{16\pi^2} |\lambda_t \mathcal{H}_2^0|^4 \left[\ln \frac{|\lambda_t \mathcal{H}_2^0|^2}{M_{st}^2} - \frac{3}{2} \right] + \text{linear terms}, \quad (28.5.45)$$

where $\lambda_t = m_t/v_2$ and $\lambda_b = m_b/v_1$ are the Yukawa couplings of the top and bottom quarks; M_{st} and M_{sb} are the masses of the stop and sbottom (the scalar superpartners of the top and bottom quarks); these masses and the terms $-3/2$ inside the square brackets are chosen to satisfy the condition that supersymmetry-breaking corrections due to stop and sbottom loops would cancel the corrections due to top and bottom loops if the masses were equal; and the ‘linear terms’ are linear in $|\mathcal{H}_2^0|^2$ or $|\mathcal{H}_1^0|^2$, with coefficients adjusted to satisfy Eq. (28.5.38). (The factor 3 takes account of the three quark colors.) Then Eq. (28.5.40) gives

$$\Delta_b = -\frac{3}{4\pi^2} |\lambda_b|^4 v_1^2 \ln \left(\frac{\lambda_b^2 v_1^2}{M_{sb}^2} \right) = \frac{3\sqrt{2} m_b^4 G_F}{2\pi^2 \cos^2 \beta} \ln \left(\frac{M_{sb}^2}{m_b^2} \right), \quad (28.5.46)$$

$$\Delta_t = -\frac{3}{4\pi^2} |\lambda_t|^4 v_2^2 \ln \left(\frac{\lambda_t^2 v_2^2}{M_{st}^2} \right) = \frac{3\sqrt{2} m_t^4 G_F}{2\pi^2 \sin^2 \beta} \ln \left(\frac{M_{st}^2}{m_t^2} \right), \quad (28.5.47)$$

where $G_F = 1.17 \times 10^{-5} \text{ GeV}^{-2}$ is the Fermi coupling constant, given by Eq. (21.3.34) as $G_F = g^2/(4\sqrt{2}m_W^2)$. Taking $m_b = 4.3 \text{ GeV}$, $m_t = 180 \text{ GeV}$, $M_{st} \sim M_{sb} \sim 1 \text{ TeV}$, and $m_Z = 91.2 \text{ GeV}$ gives $\Delta_b \sim 1.1 \times 10^{-6} m_Z^2 / \cos^2 \beta$ and $\Delta_t \sim 1.1 m_Z^2 / \sin^2 \beta$. We see that even if $\tan \beta$ is as large as m_t/m_b , the top-quark correction Δ_t will still be much larger than Δ_b .

The effect of Δ_t is to *increase* both m_H and m_h . Taking this and other radiative corrections into account [26], for $\tan \beta > 10$ the upper bound (28.5.43) on the lightest neutral

scalar mass is raised by radiative corrections from just below m_Z to between 100 GeV and 110 GeV for stop masses between 300 GeV and 1 TeV. For comparison, an experimental *lower* bound [27] of 62.5 GeV on m_h , m_H , and m_A is set by the absence of an hA or HA final state in e^+e^- collisions at 130 to 172 GeV. Also, calculations of radiative corrections involving Higgs scalars are consistent with precision measurements of electroweak phenomena for m_h in the range [27a] of 27 to 140 GeV.

Radiative corrections are less important for the charged scalars. Since the attachment of charged scalar lines allows transitions between top and bottom quarks, the correction to the scalar field potential here takes a more general form, constrained by $SU(2) \cdot U(1)$ to be

$$\Delta V = U \left(\mathcal{H}_2^\dagger \mathcal{H}_2, \mathcal{H}_1^\dagger \mathcal{H}_1, \mathcal{H}_2^\dagger \mathcal{H}_1, \mathcal{H}_1^\dagger \mathcal{H}_2, \mathcal{H}_1^T e \mathcal{H}_2 \right). \quad (28.5.48)$$

(Quark loops do not actually produce any dependence on $\mathcal{H}_1^T e \mathcal{H}_2$.) Every appearance of a $\mathcal{H}_1$ or $\mathcal{H}_2$ doublet is accompanied with a factor λ_b or λ_t , respectively, so terms involving $\mathcal{H}_1$ will be suppressed, as we have already seen in calculating the neutral scalar masses. To a good approximation, then, the correction to the effective potential is of the form

$$\Delta V \simeq U(\mathcal{H}_2^\dagger \mathcal{H}_2, 0, 0, 0) = U(|v_2 + \varphi_2|^2 + |\mathcal{H}_2^-|^2, 0, 0, 0). \quad (28.5.49)$$

By going back to the case where the charged fields vanish, we see that the function U must be just the same as what was earlier called U_t . Any terms in U of first order in an expansion in powers of $|v_2 + \varphi_2|^2 + |\mathcal{H}_2^-|^2 - v_2^2$ would simply serve to redefine the constant m_2^2 , and are eliminated by the convention (28.5.38). Terms in U of second order in $|v_2 + \varphi_2|^2 + |\mathcal{H}_2^-|^2 - v_2^2$ are genuine radiative corrections, but although they contain terms of second order in $|\varphi_2|^2$ which do affect the neutral scalar masses, they contain no terms of second order in $|\mathcal{H}_2^-|^2$ which could shift the non-Goldstone charged scalar mass. Fortunately, radiative corrections are not needed to avoid a conflict with experiment, because in the absence of an upper bound on m_A there is no theoretical upper bound on the charged scalar mass (28.5.35). An experimental lower bound [28] $m_C \geq 59$ GeV is provided by the non-observation of the process $e^+e^- \rightarrow \mathcal{H}^+\mathcal{H}^-$ at 181 to 184 GeV. There is a much more stringent lower bound on m_C provided by the rate of the process $b \rightarrow s\gamma$ (measured in decays like $B \rightarrow K^*\gamma$), which can occur by transitions to an intermediate $\mathcal{H}^-u$ or $\mathcal{H}^-c$ state, with the photon radiated from the virtual quark or $\mathcal{H}^-$. The present agreement between theory and experiment for this process sets a lower bound [29] of about 150 GeV on m_C (and higher for $\tan\beta < 1$). Using Eq. (28.5.35), this gives an important lower bound $m_A > 125$ GeV.

There are two conditions on the masses m_i^2 that must be satisfied by any model of supersymmetry breaking in order to give a successful account of electroweak symmetry breaking. One of them is provided by the requirement that the potential is bounded below, which as we have seen requires that

$$2|\mu|^2 + m_1^2 + m_2^2 > B\mu.$$

Since $B\mu$ is defined to be positive, this ensures that the C -odd neutral scalar squared mass (28.5.21) is positive. The other condition is provided by Eqs. (28.5.22) and (28.5.24), which

for arbitrary values of β require that

$$4(m_1^2 + |\mu|^2)(m_2^2 + |\mu|^2) \leq (B\mu)^2. \quad (28.5.50)$$

We can easily see from Eq. (28.5.10) that this condition ensures that the matrix of second derivatives of the potential has a negative eigenvalue at $\mathcal{H}_1 = \mathcal{H}_2 = 0$, so this $SU(2) \cdot U(1)$ -invariant point is one of unstable equilibrium, and therefore $SU(2) \cdot U(1)$ must be spontaneously broken. If β is very near $\pi/2$, then Eq. (28.5.24) tells us that this condition is satisfied by having $m_1^2 + |\mu|^2$ positive and $m_2^2 + |\mu|^2$ negative. As we will see in the next section, the renormalization group flow of the parameters in the scalar field Lagrangian provides a mechanism for driving $m_2^2 + |\mu|^2$ negative.

Even with a minimal set of superfields, in supersymmetry theories there are several pairs of particles with different $SU(2) \cdot U(1)$ transformation properties, but the same charge, color, and baryon and lepton number, which become mixed when $SU(2) \cdot U(1)$ is spontaneously broken. We have already seen an example of this in the previous section, in which we had to deal with the mixing of the scalar superpartners of the left-handed quarks with the complex conjugates of the scalar superpartners of the left-handed antiquarks. A similar mixing occurs between the higgsinos and gauginos, both charged and neutral; the particles of definite mass are not higgsinos or gauginos, but mixtures known as *charginos* and *neutralinos*. Let us consider the charginos, which provide a useful bound on μ . According to Eq. (27.4.8), there are off-diagonal supersymmetric mass terms in the Lagrangian density

$$-\text{Re} \left[\mu \left(h_{1L}^{-T} \epsilon h_{2L}^+ \right) + i\sqrt{2}m_W \cos \beta \left(w_L^{-T} \epsilon h_{2L}^+ \right) + i\sqrt{2}m_W \sin \beta \left(w_L^{+T} \epsilon h_{1L}^- \right) \right],$$

To this, we should add a term for the wino mass generated by gauge interactions with the supersymmetry-breaking sector

$$-m_{\text{wino}} \text{Re} \left(w_L^{+T} \epsilon w_L^- \right).$$

The squared chargino masses are then the eigenvalues of the matrix $\mathcal{M}_C^\dagger \mathcal{M}_C$, where

$$\mathcal{M}_C = \begin{pmatrix} m_{\text{wino}} & i\sqrt{2}m_W \sin \beta \\ i\sqrt{2}m_W \cos \beta & \mu \end{pmatrix}. \quad (28.5.51)$$

These two eigenvalues are

$$\begin{aligned} m_{\text{chargino}}^2 = & \frac{1}{2} \left[m_{\text{wino}}^2 + 2m_W^2 + |\mu|^2 \right. \\ & \pm \left((m_{\text{wino}}^2 - |\mu|^2)^2 + 4m_W^4 \cos^2 2\beta \right. \\ & \left. \left. + 4m_W^2 (m_{\text{wino}}^2 + |\mu|^2 - 2m_{\text{wino}} \text{Re} \mu \sin 2\beta) \right)^{1/2} \right]. \end{aligned} \quad (28.5.52)$$

We expect the wino mass m_{wino} to be much larger than m_W . If it is also much larger than $|\mu|$, then the heavier chargino is mostly a wino, with mass m_{wino} , while the lightest chargino is mostly a higgsino, with mass $|\mu|$. In any case, $|\mu|$ is greater than the lightest chargino mass, which the non-appearance of gauginos in e^+e^- annihilation tells us is greater than about 60 GeV, and probably greater than m_W . The search for neutralinos in e^+e^- annihilation (with assumptions like (28.4.2)) has set a lower bound of 27 GeV on the mass of the lightest neutralino [29a].

28.6 Gauge Mediation of Supersymmetry Breaking

In this section we will consider the possibility that the breakdown of supersymmetry is transmitted to the known particles through interactions of the ordinary $SU(3) \cdot SU(2) \cdot U(1)$ gauge bosons and their superpartners [30]. It is assumed here that supersymmetry is dynamically broken in a sector of superfields, *not* including the superfields of the observed quarks and leptons, and that some of the chiral superfields in the symmetry-breaking sector, known as the *messenger superfields*, have non-vanishing $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers. In order for the messenger particles to be able to get large (say, of order 1 TeV) masses without breaking $SU(3) \cdot SU(2) \cdot U(1)$, it is necessary for them to furnish a real (or pseudoreal) representation of $SU(3) \cdot SU(2) \cdot U(1)$, which automatically also means that they do not introduce any new anomalies. Although most treatments of gauge-mediated supersymmetry breaking in the literature also make specific assumptions about the interactions of the messenger superfields with the other superfields responsible for supersymmetry breaking, the most important predictions of this class of theories in fact do not depend on these assumptions. We shall therefore put off assuming anything about the interactions of the messenger superfields with the other superfields of the symmetry-breaking sector. We will, however, make another assumption about the $SU(3) \cdot SU(2) \cdot U(1)$ properties of the

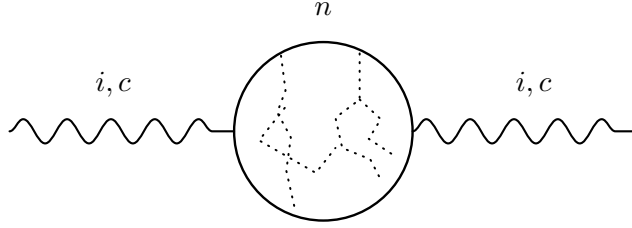


Figure 28.5. A diagram of the sort that introduces a breakdown of supersymmetry into the propagator of the gauge superfields. Here wavy lines are any component fields of the gauge superfields; solid lines are component fields of the messenger superfields; and dotted lines are component fields of the $SU(3) \cdot SU(2) \cdot U(1)$ -neutral superfields of the supersymmetry-breaking sector.

messenger superfields, which has a strong phenomenological motivation. In order for the messenger particles not to interfere with the unification of couplings discussed in Section 28.2, we assume that they have the same ratios for the total traces of all squared $SU(3) \cdot SU(2) \cdot U(1)$ gauge generators as the ordinary quarks and leptons. This condition will automatically be satisfied if the messenger superfields (together perhaps with some $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields) furnish a complete representation of some simple group G that contains $SU(3) \cdot SU(2) \cdot U(1)$, of which the quarks and leptons (again, together perhaps with some $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields) also form a complete representation. (For instance, these left-chiral superfields might form N $SU(2)$ singlet $SU(3)$ triplets with charge $e/3$ and N $SU(2)$ doublet $SU(3)$ singlets with charges 0 and $-e$, which together form N representations $\mathbf{5}$ of $SU(5)$, together with an equal number of left-chiral superfields in the complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$, which form N representations $\bar{\mathbf{5}}$.) However, for our present purposes we will neither need to

assume that G is an actual symmetry group of the theory, nor adopt any particular choice of G or of the representations furnished by the messenger particles.

This interaction of the messenger superfields both with the other chiral and/or gauge superfields of the supersymmetry-breaking sector and with the $SU(3) \cdot SU(2) \cdot U(1)$ gauge superfields can be expected to produce a breakdown of supersymmetry in the propagators of the component fields of the $SU(3) \cdot SU(2) \cdot U(1)$ gauge superfields. To lowest order in the $SU(3) \cdot SU(2) \cdot U(1)$ couplings, the leading contribution to the propagators comes from the diagram shown in Figure 28.5, in which a pair of gauge, gaugino, or auxiliary D -field lines is attached to a loop of messenger fields, which also may have any number of interactions with $SU(3) \cdot SU(2) \cdot U(1)$ -neutral fields of the supersymmetry-breaking sector. The supersymmetry-breaking corrections Δ_{ic} to the propagators of the gauge superfields (with $i = 1, 2, 3$ for $SU(3)$, $SU(2)$, and $U(1)$, and $c = V, \lambda, D$ labelling the different components of each gauge superfield) therefore have the forms

$$\begin{aligned}\Delta_{3c}(q) &= (g_s^2/16\pi^2) \sum_n T_{3n} \Pi_{cn}(q), \\ \Delta_{2c}(q) &= (g^2/16\pi^2) \sum_n T_{2n} \Pi_{cn}(q), \\ \Delta_{1c}(q) &= (g'^2/16\pi^2) \sum_n T_{1n} \Pi_{cn}(q),\end{aligned}\tag{28.6.1}$$

where n labels the different messenger superfields; $\Pi_{cn}(q)$ are more-or-less complicated functions of the four-momentum q ; T_{3n} and T_{2n} are the traces of the squares of any generator of $SU(3)$ and $SU(2)$, respectively, in the representations furnished by the n th messenger superfield (normalized so that in the defining representations $T_3 = T_2 = 1/2$); and T_{1n} is the sum of the squares of the electroweak hypercharges of the n th messenger superfield. One immediate consequence is that the gauginos acquire masses of the same form:⁴

$$\begin{aligned}m_{\text{gluino}} &= (g_s^2/16\pi^2) \sum_n T_{3n} M_{gn}, \\ m_{\text{wino}} &= (g^2/16\pi^2) \sum_n T_{2n} M_{gn}, \\ m_{\text{bino}} &= (g'^2/16\pi^2) \sum_n T_{1n} M_{gn},\end{aligned}\tag{28.6.2}$$

where the M_{gn} are masses that characterize the different messenger superfields. As already mentioned, in order to preserve the unification of couplings at very high energy we assume that the sums of the T_n have the same ratios as for the observed quarks and leptons:

$$\sum_n T_{3n} = \sum_n T_{2n} = \sum_n 3T_{1n}/5 = T.\tag{28.6.3}$$

⁴Recall that the bino is the superpartner of the $U(1)$ gauge field B_μ that appears in the Lagrangian of the standard model. We are not yet taking $SU(2) \cdot U(1)$ breaking into account, so the gaugino, squark, and slepton masses calculated here should be understood as parameters appearing in the $SU(3) \cdot SU(2) \cdot U(1)$ -invariant effective Lagrangian of the standard mode.

The breakdown of supersymmetry in these propagators is then communicated to the squarks and sleptons of the supersymmetric standard model through the diagrams shown in Figure 28.6, in which an

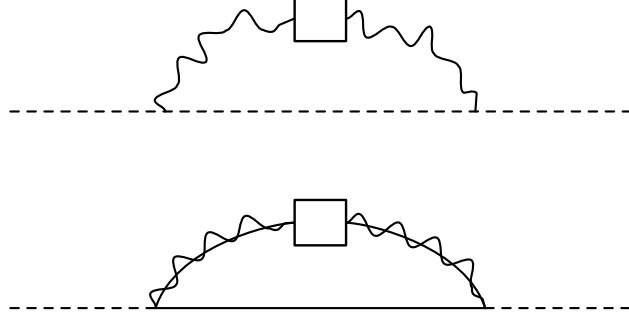


Figure 28.6. Diagrams that communicate supersymmetry breaking to the squarks and sleptons. Here dashed lines are squarks or sleptons; wavy lines are $SU(3) \cdot SU(2) \cdot U(1)$ gauge bosons or auxiliary D fields; solid lines are quarks or leptons; combined solid and wavy lines are $SU(3) \cdot SU(2) \cdot U(1)$ gauginos; and the squares represent insertions of the supersymmetry-breaking propagator correction shown in Figure 28.5.

$SU(3) \cdot SU(2) \cdot U(1)$ gauge boson or gaugino or auxiliary D field is emitted and re-absorbed by the squark or slepton. We are calculating the effective low-energy theory in which $SU(3) \cdot SU(2) \cdot U(1)$ -breaking is not yet taken into account, so there is no mixing among the $SU(3)$, $SU(2)$, and $U(1)$ propagators, and each propagator acts like a unit matrix on the gauge indices. Thus the squared mass given to any squark or slepton will be proportional to a sum over the squares of all the $SU(3) \cdot SU(2) \cdot U(1)$ generators (including coupling constants) in the representation furnished by that squark or slepton. The sums of the squared $SU(2)$ and $SU(3)$ generators in the defining representations are

$$\sum_{a=1}^3 (g\sigma_a/2)^2 = \frac{3g^2}{4} \mathbf{1}, \quad \sum_{\alpha=1}^8 (g_s\lambda_\alpha/2)^2 = \frac{4g_s^2}{3} \mathbf{1},$$

where σ_a are the Pauli isospin matrices (5.4.18) and λ_α are the Gell-Mann matrices (19.7.2). For $U(1)$ the generator is just the weak hypercharge (21.3.7), including a factor g' . The

squared squark and slepton masses therefore have the forms

$$\begin{aligned}
M_Q^2 &= 2 \sum_n M_{sn}^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 T_{3n} + \frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 T_{2n} + \left(\frac{1}{6} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_{\bar{U}}^2 &= 2 \sum_n M_{sn}^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 T_{3n} + \left(\frac{2}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_{\bar{D}}^2 &= 2 \sum_n M_{sn}^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 T_{3n} + \left(-\frac{1}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_L^2 &= 2 \sum_n M_{sn}^2 \left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 T_{2n} + \left(\frac{1}{2} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right], \\
M_{\bar{E}}^2 &= 2 \sum_n M_{sn}^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n},
\end{aligned} \tag{28.6.4}$$

where Q , $\bar{U}$, $\bar{D}$, L , and $\bar{E}$ are the scalar superpartners of the left-handed quark doublets, the left-handed antiquarks of charge $-2e/3$ and $+e/3$, the left-handed lepton doublets, and the left-handed charged antileptons, and M_{sn} are some new masses that characterize the n th messenger superfields. (The factor 2 is extracted from M_{sn}^2 for future convenience.) The squark and slepton masses that are produced in this way are automatically the same in all three generations, thus avoiding the problem with flavor-changing processes discussed in Section 28.4.

We expect all M_{gn} and M_{sn} to be roughly of the same order of magnitude, so that the gluino and squarks will have comparable masses, while the wino, bino, and sleptons will be much lighter, with masses suppressed by squares of electroweak coupling constants.

We can go considerably further than this with some reasonable dynamical assumptions. Suppose that the effects of supersymmetry breaking on the messenger superfields may be modeled by including these superfields along with a set of $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields S_n (not necessarily all distinct) in a superpotential

$$f(\Phi, \bar{\Phi}, S) = \sum_n \lambda_n S_n \Phi_n \bar{\Phi}_n, \tag{28.6.5}$$

where $\bar{\Phi}_n$ and Φ_n are left-chiral messenger superfields in complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$ and λ_n are a set of coupling coefficients. (Here and below we suppress the $SU(3) \cdot SU(2)$ indices that are summed over in calculating scalar products like $\Phi_n \bar{\Phi}_n$.) The superfields S_n are supposed to have non-vanishing vacuum expectation values $\mathcal{S}_n$ and $\mathcal{F}_n$ for their scalar and auxiliary components, respectively. It is the non-zero values of $\mathcal{F}_n$ that in these models introduce supersymmetry breaking in the masses of the Φ_n and $\bar{\Phi}_n$ particles. Section 26.4 shows that, with gauge couplings neglected, the squared masses of the spinor components of Φ_n (and $\bar{\Phi}_n$) are the eigenvalues of the matrix $\mathcal{M}_n^\dagger \mathcal{M}_n$, with $\mathcal{M}_n$ defined by Eq. (26.4.11), which gives

$$\mathcal{M}_n = \begin{pmatrix} 0 & \lambda_n \mathcal{S}_n \\ \lambda_n \mathcal{S}_n & 0 \end{pmatrix},$$

so that the messenger fermions have masses $|\lambda_n \mathcal{S}_n|$. To find the mass terms for the scalar components ϕ_n and $\bar{\phi}_n$ of the Φ_n and $\bar{\Phi}_n$ superfields, we note that integrating out the auxiliary fields of the Φ_n and $\bar{\Phi}_n$ yields a potential

$$\sum_n \left| \frac{\partial f(\phi, \bar{\phi}, \mathcal{S})}{\partial \phi_n} \right|^2 + \sum_n \left| \frac{\partial f(\phi, \bar{\phi}, \mathcal{S})}{\partial \bar{\phi}_n} \right|^2 = \sum_n |\lambda_n \mathcal{S}_n|^2 [|\phi_n|^2 + |\bar{\phi}_n|^2],$$

to which we now must add the contribution of the auxiliary component of the S_n , given by the second term of Eq. (26.4.4) as:

$$2 \operatorname{Re} \sum_n \left[\lambda_n \mathcal{F}_n \frac{\partial f(\phi, \bar{\phi}, \mathcal{S})}{\partial \mathcal{S}_n} \right] = 2 \operatorname{Re} \sum_n [\mathcal{F}_n \lambda_n \phi_n \bar{\phi}_n].$$

The complex scalar fields of definite mass are then $(\phi_n \pm e^{-i\alpha_n} \bar{\phi}_n)/\sqrt{2}$, where α_n is the phase of $\lambda_n \mathcal{F}_n$, with squared masses $|\lambda_n \mathcal{S}_n|^2 \pm |\lambda_n \mathcal{F}_n|$. (Note that this pattern, of a pair of complex scalars with squared masses equidistant above and below a Majorana fermion squared mass, is just what we would expect from the sum rule (27.5.11).) Since these squared masses must be positive, it follows that

$$|\mathcal{F}_n| \leq |\lambda_n| |\mathcal{S}_n|^2. \quad (28.6.6)$$

The gaugino masses in models based on Eq. (28.6.5) are given by diagrams of the form shown in Figure 28.5, but now with just a single loop, not including what are shown as dotted lines in Figure 28.5. A detailed calculation gives the coefficients M_{gn} in Eq. (28.6.2) as [31]

$$M_{gn} = \frac{|\mathcal{F}_n|}{|\mathcal{S}_n|} g \left(\frac{|\mathcal{F}_n|}{|\lambda_n| |\mathcal{S}_n|^2} \right), \quad (28.6.7)$$

where

$$\begin{aligned} g(x) &\equiv \frac{1}{2x^2} [(1+x) \ln(1+x) + (1-x) \ln(1-x)] \\ &= 1 + \frac{x^2}{6} + \frac{x^4}{15} + \frac{x^6}{28} + \dots \end{aligned} \quad (28.6.8)$$

The masses of the squarks and sleptons are given by the diagrams of Figure 28.6, which now involve just two loops. Another detailed calculation gives the mass parameters M_{sn}^2 in Eq. (28.6.4) as [31]

$$M_{sn}^2 = \frac{|\mathcal{F}_n|^2}{|\mathcal{S}_n^2|} f \left(\frac{|\mathcal{F}_n|}{|\lambda_n| |\mathcal{S}_n|^2} \right), \quad (28.6.9)$$

where

$$\begin{aligned} f(x) &= \frac{1+x}{x^2} \left[\ln(1+x) - 2 \operatorname{Li}_2 \left(\frac{x}{1+x} \right) + \frac{1}{2} \operatorname{Li}_2 \left(\frac{2x}{1+x} \right) \right] + x \longrightarrow -x \\ &= 1 + \frac{1}{36} x^2 - \frac{11}{450} x^4 - \frac{319}{11760} x^6 + \dots, \end{aligned} \quad (28.6.10)$$

with Li_2 the dilogarithm

$$\operatorname{Li}_2(x) \equiv - \int_0^x \frac{\ln(1-t)}{t} dt. \quad (28.6.11)$$

In particular, if (as is usually assumed) the various S_n are all the same, and if $|\mathcal{F}| \ll |\lambda_n||\mathcal{S}|^2$ for all n , then f and g in Eqs. (28.6.9) and (28.6.7) may be set equal to 1, so that

$$M_{gn} = M_{sn} = |\mathcal{F}|/|\mathcal{S}| \equiv M. \quad (28.6.12)$$

Using Eq. (28.6.3), we can express the gaugino masses (28.6.2) as

$$\begin{aligned} m_{\text{wino}} &= (g^2/16\pi^2)TM, \\ m_{\text{bino}} &= (5/3)(g'^2/16\pi^2)TM, \\ m_{\text{gluino}} &= (g_s^2/16\pi^2)TM, \end{aligned} \quad (28.6.13)$$

while the squared squark and slepton masses (28.6.4) become

$$\begin{aligned} M_Q^2 &= 2TM^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 + \frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(\frac{1}{6} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_U^2 &= 2TM^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(\frac{2}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_D^2 &= 2TM^2 \left[\frac{4}{3} \left(\frac{g_s^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(-\frac{1}{3} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_L^2 &= 2TM^2 \left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 + \frac{5}{3} \left(\frac{1}{2} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 \right], \\ M_E^2 &= 2TM^2 \frac{5}{3} \left(\frac{g'^2}{16\pi^2} \right)^2. \end{aligned} \quad (28.6.14)$$

There is no special reason to expect that $|\mathcal{F}| \ll |\lambda_n\mathcal{S}|^2$, but this assumption is not actually very restrictive, because Eq. (28.6.6) already requires that $|\mathcal{F}| \leq |\lambda_n||\mathcal{S}|^2$, and it turns out that the functions $f(x)$ and $g(x)$ do not differ much from unity for $x < 1$ unless x is very close to 1.

The extreme simplicity of the results (28.6.13) and (28.6.14) has been explained by Giudice and Rattazzi [32] by using Seiberg's arguments of holomorphy [33], described in Section 27.6. Suppose we introduce a superpotential for the messenger superfields like (28.6.5), but with a single external singlet superfield S :

$$f(S, \Phi) = S \sum_n \lambda_n \Phi_n \bar{\Phi}_n. \quad (28.6.15)$$

Among other effects the kinematic term for the gauge superfield V_i (with $i = 3, 2, 1$ for $SU(3)$, $SU(2)$, and $U(1)$) in the Wilsonian effective Lagrangian at a renormalization scale μ will now take the form

$$\mathcal{L}_{\text{gauge}, \mu} = \text{Re} \left[\sum_i N_i(S, \mu) \sum_{\alpha\beta} (W_{iL\alpha} \epsilon_{\alpha\beta} W_{iL\beta}) \right]_{\mathcal{F}}, \quad (28.6.16)$$

with some functions $N_i(S, \mu)$ replacing the factors $1/2g_i^2(\mu)$ in Eq. (27.3.22). (The θ -term is dropped here because it has no effect in perturbation theory. A sum is implied over indices

on $W_{iL\alpha}$ that label different members of the adjoint representations of $SU(3)$ and $SU(2)$, which are not explicitly shown.) The gauge coupling constant is now given by setting the superfield S equal to the expectation value $\mathcal{S}$ of its scalar component

$$\frac{1}{2g_i^2(\mu)} = N_i(\mathcal{S}, \mu). \quad (28.6.17)$$

Also, recalling that $W_{iL\alpha} = \lambda_{iL\alpha} + O(\theta)$ and using Eq. (27.2.11), the terms in the Lagrangian density (28.6.16) of second order in the gaugino fields are, up to derivative terms,

$$-2 \sum_i \text{Re} [N_i(\mathcal{S}, \mu) (\bar{\lambda}_{iR} \not{\partial} \lambda_{iR}) + [N_i(S, \mu)]_{\mathcal{F}} (\lambda_{iL}^T \epsilon \lambda_{iL})].$$

This gives the gaugino masses

$$m_{gi}(\mu) = \left| \frac{[N_i(S, \mu)]_{\mathcal{F}}}{2N_i(\mathcal{S}, \mu)} \right| = g_i^2(\mu) |[N_i(S, \mu)]_{\mathcal{F}}|. \quad (28.6.18)$$

Now let us consider the behavior of $N_i(\mathcal{S}, \mu)$ as a function of a real positive $\mathcal{S}$, with the phases of the messenger superfields adjusted so that all λ_n are real and positive. Suppose we fix the values of the gauge couplings $g_i(\mu)$ at some scale $\mu = K$ above all the messenger particle masses. Taking account of the change in the constants b_i in the renormalization group equations $\mu dg_i(\mu)/d\mu = b_i g_i^3$ as μ passes through the various messenger masses, the solution of this equation when μ is *below* all the messenger particle masses takes the form

$$\frac{1}{g_i^2(\mu)} = \frac{1}{g_i^2(K)} - 2b_i^{(0)} \ln \left(\frac{M_1}{K} \right) - 2b_i^{(1)} \ln \left(\frac{M_2}{M_1} \right) - \dots - 2b_i^{(N)} \ln \left(\frac{\mu}{M_N} \right),$$

where we label the messenger particles so that their masses $M_n = \lambda_n \mathcal{S}$ satisfy

$$M_1 > M_2 > \dots > M_N,$$

and $b_i^{(n)}$ is calculated taking account only of particles with masses less than M_n . Since all M_n are proportional to $\mathcal{S}$, we see that $N_i(\mathcal{S}, \mu)$ has the $\mathcal{S}$ dependence

$$N_i(\mathcal{S}, \mu) = -b_i^{\text{messenger}} \ln \mathcal{S} + \mathcal{S}\text{-independent terms}, \quad (28.6.19)$$

where $b_i^{\text{messenger}} = b_i^{(0)} - b_i^{(N)}$ is the contribution to b_i of all the messenger superfields. According to Eq. (27.9.45) (with $C_{i1} = 0$ and $C_{i2}^f = C_{i2}^s = \sum_n T_{in}$), this is

$$b_i^{\text{messenger}} = \frac{1}{16\pi^2} \sum_n T_{in}. \quad (28.6.20)$$

Since $N_i(S, \mu)$ is required by supersymmetry to be a holomorphic function of S , we see that

$$N_i(S, \mu) = -\frac{1}{16\pi^2} \sum_n T_{in} \ln S + S\text{-independent terms}. \quad (28.6.21)$$

Expanding around $S = \mathcal{S}$, to first order in $\mathcal{F}$ we have $[\ln S]_{\mathcal{F}} = \mathcal{F}/\mathcal{S}$ and so Eq. (28.6.18) gives the gaugino masses

$$m_{gi}(\mu) = \frac{g_i^2(\mu)}{16\pi^2} \sum_n T_{in} \left| \frac{\mathcal{F}}{\mathcal{S}} \right|. \quad (28.6.22)$$

Using Eq. (28.6.3), we see that this is the same as our previous result (28.6.13). Eq. (28.6.14) for the squark and slepton masses was obtained by Giudice and Rattazzi in a similar way, by studying the kinematic terms for the quark and lepton superfields instead of the gauge superfields.

Incidentally, Eqs. (28.6.13) and (28.6.14) could be obtained (in general with different values of M in Eq. (28.6.13) and Eq. (28.6.14)) without a specific dynamical assumption like Eq. (28.6.5), if we supposed that the supersymmetry-breaking sector respected invariance under some grand unified group G that had both the known quarks and leptons and the messenger fields as complete representations. In this case the coefficients M_{gn} and M_{sn} in Eqs. (28.6.2) and (28.6.4) would have values $M_g(d)$ and $M_s(d)$, respectively, that depend only on the irreducible representation d of G to which the n th messenger fields belong. The sums of T_{in} over n belonging to any irreducible representation d of G have the same ratios as the sums in Eq. (28.6.3), so $\sum_{n \in d} T_{in} = k_i T(d)$, where $k_3 = k_2 = 1$, $k_1 = 5/3$, and therefore

$$\sum_n T_{in} M_{gn} = \sum_d M_g(d) \sum_{n \in d} T_{in} = k_i M_g,$$

where $M_g = \sum_d M_g(d) T(d)$. Likewise

$$\sum_n T_{in} M_{sn}^2 = \sum_d M_s^2(d) \sum_{n \in d} T_{in} = k_i M_s^2,$$

where $M_s^2 = \sum_d M_s^2(d) T(d)$. Eqs. (28.6.2) and (28.6.4) would then yield Eqs. (28.6.13) and (28.6.14), except with M_g in place of TM and M_s^2 in place of $2TM^2$. But the assumption that M_{gn} and M_{sn} respect invariance under G is implausible if the messenger mass scale is far below the grand unified scale, since whatever we assume about a grand unified gauge group, the $SU(3) \cdot SU(2) \cdot U(1)$ gauge interactions would make coupling constants like λ_n run differently for Φ_n with different $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers in the same representation of the grand unified group.

These results are subject to various radiative corrections, of which the most important is that we must use values of g_s , g , and g' renormalized at a scale comparable to the mass being calculated. Indeed, the gaugino mass ratios given in Eq. (28.6.13) could also be derived under quite different assumptions: that all gaugino masses are equal at the grand unification scale where the coupling constants are related by $g_s^2 = g^2 = 5g'^2/3$ and become different at lower energies as described by the renormalization group equations.

For an illustrative example of numerical results, suppose that the messenger superfields form an $SU(2)$ singlet $SU(3)$ triplet with charge $e/3$ and an $SU(2)$ doublet $SU(3)$ singlet with charges 0 and $-e$, together with left-chiral superfields in the complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$. Then, as already mentioned, Eq. (28.6.3) is satisfied with $T = 2 \cdot 1/2 = 1$, so by using the correct values of the gauge couplings, the masses of the squarks, gluino, L sleptons, wino, E sleptons, and bino are calculated to be in the ratios [34] $11.6 :: 7.0 :: 2.5 :: 2 :: 1.1 :: 1.0$. In larger representations of G we could have $T \gg 1$, in which case the gluino would be the heaviest of these particles, and the sleptons the lightest ones.

There are also radiative corrections aside from the running of the gauge couplings. According to one calculation [34], in the model with an $SU(2)$ singlet $SU(3)$ triplet with charge $e/3$ and an $SU(2)$ doublet $SU(3)$ singlet with charges 0 and $-e$, together with left-chiral superfields in the complex-conjugate representations of $SU(3) \cdot SU(2) \cdot U(1)$, radiative corrections give the ratios of the masses of the squarks, gluino, L sleptons, wino, E sleptons, and bino as $9.3 :: 6.4 :: 2.6 :: 1.9 :: 1.35 :: 1.0$.

As we saw in the previous section, the wino and bino may mix with the charged and neutral higgsinos, so the wino and bino masses that have been calculated here must be regarded as inputs to a calculation of physical masses of mixtures known as *charginos* and *neutralinos*, rather than physical masses themselves.

Now we shall consider the masses of the Higgs scalars in these models. If we were to consider only the two-loop diagrams by which these scalars get masses through gauge interactions with the supersymmetry-breaking sector, then since (apart from signs) they have the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers as the left-handed lepton doublets, their masses would be given by a formula like the fourth of Eqs. (28.6.4)

$$\begin{aligned} [m_1^2]_{2 \text{ loop}} &= [m_2^2]_{2 \text{ loop}} = M_L^2 \\ &= \sum_n M_{sn}^2 \left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 T_{2n} + \left(\frac{1}{2} \right)^2 \left(\frac{g'^2}{16\pi^2} \right)^2 T_{1n} \right]^2. \end{aligned} \quad (28.6.23)$$

If this were the whole story, then it would be impossible to meet the condition for $SU(2) \cdot U(1)$ breaking found in the previous section that (unless $\tan \beta$ is very close to unity) one of $m_1^2 + |\mu|^2$ and $m_2^2 + |\mu|^2$ must be negative. Fortunately, the large masses of the top quark and squark produce a negative contribution to m_2^2 that leads naturally to a spontaneous breakdown of electroweak symmetry. The couplings of the Higgs doublets to the third-generation quark superfields are described by the superpotential

$$f_{3\text{rd gen}} = \lambda_b (H_1^T e Q) \bar{B} + \lambda_t (H_2^T e Q) \bar{T}, \quad (28.6.24)$$

where Q is the $SU(2)$ quark doublet left-chiral superfield (T, B) , while $\bar{T}$ and $\bar{B}$ are the left-chiral superfields of the left-handed top and bottom antiquarks, and λ_t and λ_b are the Yukawa couplings, related to the t and b quark masses by $m_t = \lambda_t v_2$ and $m_b = \lambda_b v_1$. The last term of Eq. (26.4.7) then gives the terms in the potential involving an interaction between the squark fields and Higgs fields as

$$\begin{aligned} V_{\text{sq } H} &= |\lambda_b \mathcal{H}_1^- \bar{\mathcal{B}} + \lambda_t \mathcal{H}_2^0 \bar{\mathcal{T}}|^2 + |\lambda_b \mathcal{H}_1^0 \bar{\mathcal{B}} + \lambda_t \mathcal{H}_2^+ \bar{\mathcal{T}}|^2 \\ &\quad + |\lambda_b|^2 |\mathcal{H}_1^0 \mathcal{B} - \mathcal{H}_1^- \mathcal{T}|^2 + |\lambda_t|^2 |\mathcal{H}_2^+ \mathcal{B} - \mathcal{H}_2^0 \mathcal{T}|^2. \end{aligned} \quad (28.6.25)$$

where script letters denote the scalar field components of superfields. The squark loop contributions to the potential of $\mathcal{H}_1$ and $\mathcal{H}_2$ are then

$$V_H^{\text{squark loop}} = 3 \langle \mathcal{S} \mathcal{S}^* \rangle \left[2 |\lambda_b|^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + 2 |\lambda_t|^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) \right], \quad (28.6.26)$$

where $\langle \mathcal{S} \mathcal{S}^* \rangle$ is the vacuum expectation value of the product of any one of the squark fields and its complex conjugate at the same spacetime point. (In taking this to be the same

for all squark types, we are here using Eq. (28.6.4), which tells us that the squark masses M_Q do not vary much among $\mathcal{T}$, $\mathcal{B}$, $\bar{\mathcal{T}}$, and $\bar{\mathcal{B}}$ squarks. The factor 3 in Eq. (28.6.26) takes account of the three colors of each squark type.) The vacuum expectation value $\langle \mathcal{S}\mathcal{S}^* \rangle$ is given in lowest order by

$$\langle \mathcal{S}\mathcal{S}^* \rangle \equiv \langle \mathcal{S}(x)\mathcal{S}^*(x) \rangle_{\text{VAC}} = \frac{-i}{(2\pi)^4} \int \frac{d^4p}{p^2 + M_Q^2 - i\epsilon}. \quad (28.6.27)$$

This of course is divergent, but the contribution to supersymmetry-breaking coefficients would be cancelled by quark loops if the squarks had the zero bare mass of the quarks, so the effect of quark loops is to subtract from (28.6.27) the same expression with M_Q replaced with zero, which with a Wick rotation becomes

$$\begin{aligned} \langle \mathcal{S}\mathcal{S}^* \rangle &\longrightarrow \frac{M_Q^2 i}{(2\pi)^4} \int \frac{d^4p}{(p^2 + M_Q^2 - i\epsilon)(p^2 - i\epsilon)} \\ &= -\frac{M_Q^2}{(2\pi)^4} \int_0^{M^2} \frac{\pi^2 dp^2}{p^2 + M_Q^2} \simeq -\frac{M_Q^2}{16\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right). \end{aligned}$$

We have inserted an ultraviolet cut-off at the messenger mass M , because at momenta above M the squark mass must be replaced with a momentum-dependent mass, which goes to the supersymmetric value zero at very high momenta. Making this substitution in Eq. (28.6.26) gives a net contribution to the potential due to squark and quark loops:

$$(V_m)^{3 \text{ loop}} = \frac{3M_Q^2}{16\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right) \left[2|\lambda_b|^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + 2|\lambda_t|^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]. \quad (28.6.28)$$

(This is a three-loop contribution, because the squared squark masses are given by two-loop diagrams. There are also terms in the potential that are quadratic in both the Higgs and squark superfields, arising from products of the Higgs and squark terms in the squares of the $SU(2) \cdot U(1)$ gauge field D -components. These do not make a three-loop contribution to the Higgs masses because the sum of each $SU(2) \cdot U(1)$ quantum number of the squarks vanishes, so that their contributions to the $SU(2) \cdot U(1)$ gauge field D -components have zero vacuum expectation value.) Comparing Eq. (28.6.28) with Eq. (28.5.9) and adding the two-loop contribution (28.6.23) to the masses, we see then that

$$m_1^2 \simeq M_L^2 - \frac{3M_Q^2 |\lambda_b|^2}{8\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right), \quad (28.6.29)$$

$$m_2^2 \simeq M_L^2 - \frac{3M_Q^2 |\lambda_t|^2}{8\pi^2} \ln \left(\frac{M^2}{M_Q^2} \right). \quad (28.6.30)$$

Using Eq. (28.6.14) and $|\lambda_t| = m_t/v_2 = m_t(2\sqrt{2}G_F)^{1/2}/\sin\beta$, we may write (28.6.30) as

$$\begin{aligned} m_2^2 \simeq 2TM^2 &\left[\frac{3}{4} \left(\frac{g^2}{16\pi^2} \right)^2 + \frac{5}{12} \left(\frac{g'^2}{16\pi^2} \right)^2 \right. \\ &\left. - \frac{\sqrt{2}G_F m_t^2}{\pi^2 \sin^2 \beta} \left(\frac{g_s^2}{16\pi^2} \right)^2 \ln \left(\frac{3}{8T(g_s^2/16\pi^2)} \right) \right]. \end{aligned} \quad (28.6.31)$$

For $T = 1$, $g_s^2/4\pi = 0.118$, $g^2/4\pi = 0.0340$, $g'^2/4\pi = 0.0101$, and $m_t = 180$ GeV, this is

$$m_2^2 \simeq M_L^2 \left[1 - \frac{3.06}{\sin^2 \beta} \right], \quad (28.6.32)$$

which is negative for all values of β , thus providing a natural mechanism for the spontaneous breaking of the electroweak gauge symmetry. Also, $M_L^2 = (0.91 \cdot 10^{-4}) M^2 / 8\pi^2$. Unless $\tan \beta$ is huge, we have $|\lambda_b| \ll |\lambda_t|$, so Eq. (28.6.27) gives

$$m_1^2 \simeq M_L^2. \quad (28.6.33)$$

The aspect of electroweak phenomenology for which the predictions of gauge-mediated supersymmetry models are most uncertain and unsatisfactory has to do with the parameter μ in the supersymmetry-preserving term $\mu[(H_1^T e H_2)]_{\mathcal{F}}$ and the related supersymmetry-breaking term $B\mu$ in the Lagrangian density. These are related because the interactions of the Higgs superfields with the gauge, lepton, and quark superfields are invariant under a symmetry

$$\begin{aligned} H_1 &\longrightarrow e^{i\varphi} H_1, & H_2 &\longrightarrow e^{i\varphi} H_2, \\ Q &\longrightarrow e^{-i\varphi} Q, & V_i &\longrightarrow V_i, \\ \bar{D} &\longrightarrow \bar{D}, & \bar{U} &\longrightarrow \bar{U}, \end{aligned} \quad (28.6.34)$$

which in the absence of a superpotential term $\mu(H_1^T e H_2)$ would forbid radiative corrections from producing a term $B\mu \text{Re}(\mathcal{H}_1^T e \mathcal{H}_2)$ in the scalar field potential.

It is not possible for $B\mu$ to vanish because then Eq. (28.5.22) and the fact that (as we have seen) $m_A \neq 0$ would imply that $\sin 2\beta = 0$, or in other words either $v_1 = 0$ or $v_2 = 0$, which would imply that either all the charge $-e/3$ quarks and charged leptons are massless or all the charge $+2e/3$ quarks are massless. (If $B\mu = 0$ and $\mu = 0$ then this problem persists to all orders, because the appearance of non-zero vacuum expectation values for both v_1 and v_2 would imply that symmetry under any combination of the transformation (28.6.34) and electroweak $U(1)$ gauge transformations is spontaneously broken, so the C -odd neutral scalar would be a Goldstone boson with $m_A = 0$.) It is natural to try to account for a non-zero value of $B\mu$ as a radiative correction in a theory in which symmetry under the transformation (28.6.34) is explicitly broken by a supersymmetric term $\mu[(H_1^T e H_2)]_{\mathcal{F}}$ in the Lagrangian density. This gives a very small value of $B\mu$ at the messenger scale [34], although renormalization group effects greatly increase $B\mu$ at lower energies. According to Eq. (28.5.22), a relatively small $B\mu$ would fit well with the idea that the large mass of the top quark arises from a large value of $\tan \beta$. In any case, as already mentioned, the experimental lower bound on the chargino mass tells us that $|\mu|$ is at least about 60 GeV.

The trouble is that the appearance of a non-zero value of μ resurrects the hierarchy problem that supersymmetry was supposed to solve: instead of asking why the Higgs mass terms in the Lagrangian density are so much smaller than the Planck mass or the mass at which gauge couplings are unified, we now have to ask why μ is this small?

The hierarchy problem would be put to rest if the Higgs superfields interact with the supersymmetry-breaking sector in such a way that a term $\mu[(H_1^T e H_2)]_{\mathcal{F}}$ is forbidden by some symmetry, but appears when that symmetry is spontaneously broken. A massless

Goldstone boson may be avoided if the symmetry is discrete rather than continuous. The simplest possibility is simply to extend the symmetry transformation (28.6.34) to include a transformation

$$S \longrightarrow e^{-2i\varphi} S,$$

which would allow a term in the superpotential of the form

$$\lambda' S (\mathcal{H}_1^T e \mathcal{H}_2).$$

We can avoid a continuous symmetry by also including a term S^3 in the superpotential, so that the Lagrangian is invariant only under transformations with φ a multiple of $2\pi/3$, which is still enough to forbid a non-zero bare value of μ . In this case the appearance of non-zero vacuum values $\mathcal{S}$ and $\mathcal{F}$ for the scalar and auxiliary components of S yields

$$B\mu = |\lambda' \mathcal{F}|, \quad \mu = |\lambda' \mathcal{S}|.$$

This would result in B having a very large $|\lambda'|$ -independent value M , given by (28.6.12), which is greater by a factor of order $(g_s^2/16\pi)^{-1} \simeq 100$ than the squark or gluino masses. But then, since Eqs. (28.6.32) and (28.6.33) give $m_1^2 + m_2^2 < 0$, the stability condition (28.5.11) would require that $|\mu| \geq M/2$, and hence Eq. (28.5.22) would require that m_A is also much greater than the squark and slepton masses. This is ruled out by the relation (28.5.23) and the estimates (28.6.33) and (28.6.32) of m_1^2 and m_2^2 unless $\tan \beta$ is very close to unity.

We will see in Section 31.6 that theories of gravity-mediated supersymmetry breaking naturally yield acceptable values of $B\mu$ and μ . Such theories are characterized by a very high energy scale of supersymmetry breaking, in various versions of order either 10^{11} GeV or 10^{13} GeV. There have been several suggestions [35] on ways to obtain acceptable values of $B\mu$ and μ in theories in which supersymmetry is broken at relatively low energy, such as theories of gauge-mediated supersymmetry breaking, but none of them is particularly compelling. Also, since we do not know where μ comes from, we do not have any reason to suppose that it is real, so theories of gauge-mediated supersymmetry breaking are at risk of yielding too much CP violation, just as in the more general framework of Section 28.4.

Like the scalar squared masses m_1^2 and m_2^2 , the parameters A_{ij} and C_{ij} in Eq. (28.4.1) are given by two-loop diagrams. However, they have the dimensions of mass rather than mass squared, and they are much less than the scalar and gaugino masses, so they make a relatively unimportant contribution to supersymmetry breaking.

As in any model with supersymmetry broken at energies much less than 10^{10} GeV, the lightest R -odd particle in all models based on gauge-mediated supersymmetry breaking is the gravitino. As we will see in Section 31.3, the gravitino mass is of the order of $\sqrt{G}$ times the squared energy F characteristic of supersymmetry breaking, defined so that the vacuum energy is $F^2/2$. Where supersymmetry is broken by the $\mathcal{F}$ -terms $\mathcal{F}_{n0}$ of the $SU(3) \cdot SU(2) \cdot U(1)$ -neutral chiral superfields S_n , we have $F^2 = \sum_n |\mathcal{F}_{n0}|^2$. If there are no large dimensionless quantities in the Lagrangian for the S_n , then squark masses in this model are of the order of $g_s^2 \sqrt{F}/16\pi^2 \simeq 10^{-2} \sqrt{F}$, so in order for this to be less than the naturalness bound of 10^4 GeV, we must have $\sqrt{F} < 10^6$ GeV, which gives a gravitino mass

less than 1 keV. Gravitational couplings are so weak at accessible energies that it is only the helicity $\pm 1/2$ states of the gravitino that can actually be produced, and these behave just like goldstino states. As shown by Eq. (27.5.12), in the models discussed here the goldstino field appears with coefficient $i\sqrt{2}\mathcal{F}_n$ in the fermionic component ψ_n of S_n . Goldstinos are emitted in the decay of the R -odd sparticles into the corresponding R -even particles of the standard model through radiative corrections, with goldstinos emerging from vertices connecting internal ψ_n and S_n lines. According to Eq. (29.2.10), the goldstino emission amplitudes are inversely proportional to F , which makes these decays comparatively slow, although perhaps fast enough to be observed.

Because the decay of R -odd particles into goldstinos is slow, it is phenomenologically important in these models to identify the next-to-lightest R -odd particle, into which all the heavier R -odd particles will decay before it in turn decays into a goldstino. As we have seen, the next-to-lightest R -odd particle is usually a slepton, wino, or bino. (It is possible in models with a messenger superfield having the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers as the Higgs doublets that the mixing between these superfields would lower the mass of the doublet messengers so much that the lightest R -odd particle would be the gluino [35a].) Detailed calculations including the effects of $SU(2) \cdot U(1)$ breaking indicate that here is a large region of parameter space in which the next-to-lightest R -odd particle is one of the two tau sleptons [35b].

28.7 Baryon and Lepton Non-Conservation

The extra particles in supersymmetric models provide several new mechanisms for baryon and lepton non-conservation. We saw in Section 28.1 that there are various baryon- and lepton-number non-conserving supersymmetric operators (28.1.2) and (28.1.3) of dimensionality four that can be included in a renormalizable $SU(3) \cdot SU(2) \cdot U(1)$ -invariant theory and that would lead to processes like proton decay at a catastrophic rate. These terms can be excluded from the Lagrangian by imposing R parity conservation (or, equivalently, invariance under a change of sign of all quark and lepton chiral superfields) but this does not exclude various $SU(3) \cdot SU(2) \cdot U(1)$ -invariant but baryon- and lepton-number non-conserving operators of dimensionality $d > 4$. As discussed in Section 21.3, if there is an underlying mechanism for baryon and lepton non-conservation characterized by some high mass scale M , then these operators will appear in the effective Lagrangian of the standard model with coefficients proportional to M^{4-d} . With only the fields of the non-supersymmetric standard model, operators that can violate baryon conservation have a minimum dimensionality six [36], and thus give baryon non-conserving amplitudes proportional to M^{-2} . The new fields required by supersymmetry lead to two important changes in estimates of baryon non-conserving processes like proton and bound neutron decay. As we saw in Section 28.2, the change in the renormalization group equations gives larger estimates of M , which decreases the effect of the dimension six operators. At the same time, these new fields allow the construction of new operators of dimensionality five, which give baryon non-conserving amplitudes proportional to M^{-1} and are therefore likely to make the dominant contribution to proton and bound neutron decay [37].

The supersymmetric operators of dimensionality five that can be formed out of chiral superfields (generically called Φ) are of the forms

$$(\Phi^* \Phi \Phi)_D \quad \text{and} \quad (\Phi \Phi \Phi \Phi)_{\mathcal{F}},$$

and their complex conjugates. (We do not consider operators that include derivatives or gauge fields, because they do not offer any additional possibility of baryon or lepton non-conservation.)

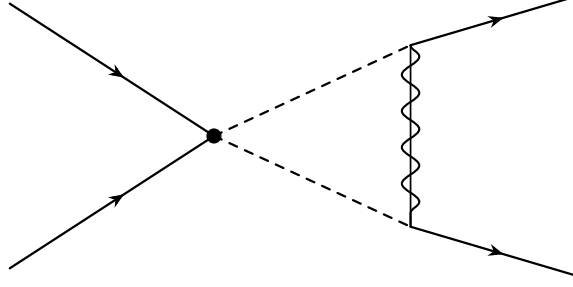


Figure 28.7. A diagram that can produce a four-fermion interaction among quarks and/or leptons that violates baryon and lepton number conservation. Here solid lines are quarks and/or leptons; dashed lines are squarks and/or sleptons; the combined solid and wavy line is a gaugino; and the dot is a vertex arising directly from the $\mathcal{F}$ -term interactions (28.7.3).

In the notation of Section 28.1, the $SU(3) \cdot SU(2) \cdot U(1)$ -invariant operators of dimensionality five that also conserve R parity are

$$(LLH_2H_2)_{\mathcal{F}}, \tag{28.7.1}$$

$$(L\bar{E}H_2^*)_D, \quad (Q\bar{D}H_2^*)_D, \quad (Q\bar{U}H_1^*)_D, \quad (QQ\bar{U}\bar{D})_{\mathcal{F}}, \quad (Q\bar{U}L\bar{E})_{\mathcal{F}}, \tag{28.7.2}$$

and

$$(QQQL)_{\mathcal{F}}, \quad (\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}, \tag{28.7.3}$$

with obvious contraction of indices as dictated by $SU(3)$ and $SU(2)$ conservation. The interaction (28.7.1) is the supersymmetric version of the dimensionality five operator which would provide small neutrino masses in some theories [38]. The interactions (28.7.2) only provide small corrections to processes that already occur in the renormalizable terms of the supersymmetric standard model. It is the interactions (28.7.3) that provide new mechanisms for baryon as well as lepton non-conservation.

According to Eq. (26.4.4), the quarks and leptons enter in the interactions (28.7.3) through terms involving a pair of quark and/or lepton fields and a pair of squark and/or slepton fields. In order to generate reactions among quarks and leptons alone, it is necessary for the pair of squarks and/or sleptons to be converted into a pair of quarks and/or leptons by exchanging a gaugino in the one-loop diagram shown in Figure 28.7. This will produce effective four-fermion $qqq\ell$ interactions with $d = 6$ among three quarks and a lepton. The coupling g_6 of these interactions will be proportional to the square of the gauge coupling g or g' or g_s of the gaugino, to the supersymmetry-breaking mass of the gaugino, to the

inverse square of the larger of the gaugino and squark or slepton masses (which is needed to give a coupling of the right dimensionality), and to a factor of order $1/8\pi^2$ from the loop integration.

It might be thought that the greater strength of the gluino coupling would make gluino exchange the dominant contribution to g_6 . (Indeed, in gauge-mediated supersymmetry-breaking theories, for moderate values of the trace (28.6.3) and with $g \approx g'$, Eqs. (28.6.13)–(28.6.14) give

$$m_{\text{gluino}} \approx m_{\text{squark}} \approx \frac{g_s^2}{16\pi^2} M_*, \quad m_{\text{wino}} \approx m_{\text{slepton}} \approx m_{\text{bino}} \approx \frac{g^2}{16\pi^2} M_*,$$

where M_* is a mass characterizing the messenger sector. Therefore in such theories individual gluino exchange diagrams give contributions proportional to g_s^2/m_{gluino} , while wino or bino exchange (or, more precisely, chargino or neutralino exchange) makes a contribution proportional to $g^2 m_{\text{wino}}/m_{\text{squark}}^2$, which is smaller by a factor roughly equal to $m_{\text{wino}} g_s^2/m_{\text{gluino}} g^2 \approx g^4/g_s^4$.) However, there is a cancellation among the diagrams for gluino exchange that strongly suppresses their contribution. This was originally shown using a Fierz identity among four-fermion operators [39], but the same result can be obtained with no equations at all. To conserve color, the coefficients of the operators $(QQQL)_{\mathcal{F}}$ and $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ must be totally antisymmetric in the colors of the three quark or antiquark superfields, and since these superfields are bosonic, they must then also be antisymmetric in their flavors as well. Gluino interactions are flavor-independent, so if we can neglect the flavor dependence of the squark masses then the coefficients of the four-fermion $d = 6$ operators will also be totally antisymmetric in flavor as well as color. Fermi statistics then requires the coefficients of these operators to be also totally antisymmetric in the spin indices of the quark or antiquark fields. But the three quark or antiquark fields in the $d = 6$ operators derived by gaugino exchange from $(QQQL)_{\mathcal{F}}$ or $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ are all left-handed and therefore have only two independent spin indices, so no coefficient can be antisymmetric in all three spins. The contribution of gluino exchange to the $d = 6$ operators would therefore vanish if the squark masses were all equal and so this contribution is suppressed by the fractional differences among the masses of the different squarks. According to Eq. (28.6.4), in theories of gauge-mediated supersymmetry breaking the fractional differences between the $\bar{U}$ and $\bar{D}$ squark masses are of order g'^4/g_s^4 , so gluino exchange between the antisquarks in the $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ operator generates a dimension six four-fermion operator with a coefficient of the same order as that generated by bino exchange. However, since gluinos conserve flavor this operator like the $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ operator must be totally antisymmetric in antiquark flavors, so that it must involve c or t quarks and therefore cannot contribute directly to proton or bound neutron decay. On the other hand, Eq. (28.6.4) indicates that the fractional mass differences among the Q squarks of different flavors is much less than of order g^4/g_s^4 , so that gluino exchange between the squarks in the $(QQQL)_{\mathcal{F}}$ operator makes a contribution to g_5 that is much smaller than wino or bino exchange. We conclude that at least in the theories of gauge-mediated supersymmetry breaking, gluino exchange makes a smaller contribution to proton or bound neutron decay than wino or bino exchange. In other models gluino exchange may make a contribution comparable to that of other processes [40].

For $g \approx g'$ and $m_{\text{wino}} \approx m_{\text{bino}}$, the contribution of wino or bino exchange to the dimension six operators is of order

$$g_6 \approx \frac{g^2 g_5 m_{\text{wino}}}{8\pi^2 m_{\text{squark}}^2}, \quad (28.7.4)$$

where g_5 is a typical value of the couplings of the effective $d = 5$ interactions (28.7.3). If the wino and squark masses have the same ratio (g^2/g_s^2) as in theories of gauge-mediated supersymmetry breaking, then this gives

$$g_6 \approx \frac{g^4 g_5}{8\pi^2 g_s^2 m_{\text{squark}}^2}. \quad (28.7.5)$$

The four-fermion $qqq\ell$ -terms of dimensionality six in the effective Lagrangian are the same as those that had been supposed to generate processes like proton decay in non-supersymmetric theories [36]. They produce proton and bound neutron decay at a rate which on dimensional grounds must be of the form

$$\Gamma_N = c_N m_N^5 g_6^2, \quad (28.7.6)$$

where c_N is a pure number that must be calculated by non-perturbative calculations in quantum chromodynamics. Much work has gone into these calculations, with results [41] generally in the range $c_N \approx 3 \times 10^{-3 \pm 0.7}$.

To estimate g_5 , we note that it is not possible to produce $\mathcal{F}$ -terms like (28.7.3) that involve only left-chiral superfields by the tree-approximation exchange of gauge supermultiplets, which always interact with both left-chiral superfields and their right-chiral complex conjugates. Thus the interactions (28.7.3) arise in the tree approximation only from the exchange of particles of chiral superfields, and therefore g_5 is of the order of g_T^2/M_T , where g_T is a typical baryon and lepton non-conserving coupling of some superheavy left-chiral superfield of mass M_T to the quark and lepton superfields. To produce the interactions (28.7.3), these superheavy particles must be color triplets or antitriplets, and $SU(2)$ triplets or singlets. Whatever gauge group unifies the strong and electroweak interactions presumably dictates some relation between the interactions of the superheavy color triplet T and the familiar color singlet H_1 and H_2 . Then g_T will be of the same order as the Yukawa coupling constants in the interactions (28.1.2) and (28.1.3) that give mass to the known quarks and leptons, and which are equal to quark or lepton masses divided by the vacuum expectation values of order $G_F^{-1/2} \simeq 300$ GeV of $\mathcal{H}_1^0$ or $\mathcal{H}_2^0$. We therefore take

$$g_5 \approx \frac{G_F m_f^2}{M_T}, \quad (28.7.7)$$

where m_f is some typical quark or lepton mass. We have seen that the dimension five operators are antisymmetric in quark flavors, so as a compromise between the masses of the s quark and the u or d quarks, we will take $m_f = 30$ MeV. Combining Eqs. (28.7.5)–(28.7.7) and taking $M_T = 2 \times 10^{16}$ GeV (as suggested by the results of Section 28.2), $c_N = 0.003$, $g_s^2/4\pi = 0.118$, $g^2/4\pi = 1/(0.23 \times 137)$, and $m_{\text{squark}} = 1$ TeV, we find a proton

(or bound neutron) lifetime Γ_N^{-1} of about 2×10^{31} years [42]. This is not very different from experimental lower bounds on the partial lifetimes of what are expected to be the leading modes of proton decay, variously quoted as ranging from 10^{31} to 5×10^{32} years. At the time of writing the most stringent bounds are set by the non-observation of proton decay in the large Super Kamiokande neutrino detector in Japan [42a]: the partial lifetimes for the decays $p \rightarrow e^+ \pi^0$ and $p \rightarrow \bar{\nu} K^+$ are greater than 2.1×10^{33} years and 5.5×10^{32} years, respectively. There is an uncertainty of a factor of at least 100 in the above estimate of the theoretical lifetime from the uncertainty in the squark mass alone, so it is too soon to say that there is any discrepancy between experiment and theoretical expectations. On the other hand, supersymmetry raises the possibility that baryon non-conservation may be discovered soon.

We can also say a little on general grounds about the expected branching ratios for various proton and bound neutron decay modes. As we have mentioned, the dimension five operators (28.7.3) must be totally antisymmetric in the flavor of the quark superfields, so the only operators that concern us here are of the forms $(U_i D_j D_k N_\ell)_\mathcal{F}$, $(D_i U_j U_k E_\ell)_\mathcal{F}$, and $(\bar{D}_i \bar{U}_j \bar{U}_k \bar{E}_\ell)_\mathcal{F}$, where i, j, k, ℓ are generation indices, and in each case $j \neq k$. The exchange of neutral winos or binos then produces $d = 6$ four-fermion operators of the forms $u_i d_j d_k \nu_\ell$, $d_i u_j u_k e_\ell$, and $\bar{d}_i \bar{u}_j \bar{u}_k \bar{e}_\ell$, with $j \neq k$ and i and ℓ arbitrary, while charged wino exchange produces the same four-fermion operators with $i \neq j$ and k and ℓ arbitrary. The only quarks light enough to be involved in proton decay are u , s , and d ; ignoring all the others and the small mixing angles in the third generation, we have

$$u_1 = u, \quad d_1 = d \cos \theta_c + s \sin \theta_c, \quad d_2 = -d \sin \theta_c + s \cos \theta_c,$$

where θ_c is the Cabibbo angle, while u_2 , u_3 , and d_3 can be ignored. The four-fermion operators that can be produced by wino or bino exchange and can contribute to proton or bound neutron decay thus are $uds\nu_\ell \cos(2\theta_c)$, $udd\nu_\ell \sin(2\theta_c)$, $uuse_\ell \cos \theta_c$, and $uude_\ell \sin \theta_c$, plus others with quarks and leptons replaced with antiquarks and antileptons. All other things being equal, the dominant decay modes are therefore $p \rightarrow K^+ \bar{\nu}$, $n \rightarrow K^0 \bar{\nu}$, $p \rightarrow K^0 e^+$, and $p \rightarrow K^0 \mu^+$, while the rates for the decay modes $p \rightarrow \pi^+ \bar{\nu}$, $n \rightarrow \pi^0 \bar{\nu}$, $p \rightarrow \pi^0 e^+$, $p \rightarrow \pi^0 \mu^+$, and $n \rightarrow \pi^- e^+$ are suppressed by factors $\sin^2 \theta_c = 0.05$, though also enhanced somewhat by the greater phase space available.

These considerations do not lead to definite predictions of branching ratios, because in addition to all the factors mentioned above, the coefficients generically called g_5 of the operators (28.7.3) may have a strong dependence on the superfield flavors appearing in these operators. To go further, one needs a specific theory for the generation of the dimension five operators. Most of the authors of Reference 42 concluded on the basis of a supersymmetric version of $SU(5)$ theories that proton and bound neutron decay would be dominated by the processes $p \rightarrow K^+ \bar{\nu}$ and $n \rightarrow K^0 \bar{\nu}$, but for a model based on $SO(10)$ charged lepton modes can become prominent [43]. Also, in some models higgsino exchange can compete with wino and bino exchange [44], increasing the rate of $p \rightarrow K^+ \bar{\nu}$. It seems a good idea in searches for baryon non-conservation to keep an open mind as to the decay modes to be expected in proton or bound neutron decay.

Of course, it is possible that all of these baryon non-conserving processes are prohibited by some sort of conservation law. As mentioned in Section 28.1, string theory argues against baryon conservation being a fundamental global continuous symmetry, but the baryon non-conserving operators (28.7.3) may be forbidden by a $\mathbb{Z}_3$ multiplicative symmetry known as baryon parity [45], under which the Q superfield is neutral; the H_2 and $\bar{D}$ superfields are multiplied by the phase $\exp(i\pi/3)$, and the L , H_1 , $\bar{U}$, and $\bar{E}$ superfields are multiplied by the opposite phase $\exp(-i\pi/3)$. This symmetry is designed to allow the fundamental Yukawa couplings (28.1.2) and (28.1.3) as well as the μ -term (28.5.7) and the lepton non-conserving terms (28.1.4) and (28.7.1), but it rules out the dimension four baryon non-conserving term (28.1.5) and the dimension five baryon non-conserving terms (28.7.3). This symmetry is spontaneously broken by the appearance of vacuum expectation values of $\mathcal{H}_1^0$ and $\mathcal{H}_2^0$ (and perhaps the sneutrino fields $\mathcal{N}$), and with no conservation law for R parity, there would be nothing to keep the lightest supersymmetric particle stable.

Problems

1. Suppose that the interactions (28.1.4) and (28.1.5) were actually present in the Lagrangian of a supersymmetric version of the standard model. Make a rough estimate of how heavy the squarks and sleptons would have to be to avoid a conflict with experimental bounds on the proton lifetime.
2. Suppose that the typical mass m of the gauginos, higgsinos, squarks and sleptons is very much larger than m_Z . Give the renormalization group equations for the running gauge couplings at energies both above and below m . Use the results, together with the unification assumption employed in Section 28.2, to give formulas for $\sin^2 \theta$ and the unification scale M in terms of m , m_Z , $e(m_Z)$, $g_s(m_Z)$, and n_s . How large could m be without violating experimental bounds on $\sin^2 \theta$ and M ?
3. Give formulas for the couplings of the quarks and leptons to the lightest CP-even neutral scalar particle in the minimum supersymmetric standard model in terms of the parameters m_A , m_Z , β , G_F , and the quark and lepton masses.
4. Use holomorphy arguments to derive the one-loop formula for the gluino mass in a theory of gauge-mediated supersymmetry breaking, in which the messenger superfields Φ_n and $\bar{\Phi}_n$ get their masses from a term $\sum_n \lambda_n S_n (\bar{\Phi}_n \Phi_n)$ in the superpotential, in terms of the expectation values $\mathcal{S}_n$ and $\mathcal{F}_n$ of the ϕ - and F -components of the singlet superfields S_n , in the limit where $|\mathcal{F}_n| \ll |\lambda_n| |\mathcal{S}_n|^2$.
5. Taking account of the possibility of a small flavor dependence of squark masses, estimate the gluino exchange contribution to baryon and lepton non-conserving four-fermion interactions among the quarks and leptons. Set an upper bound on these contributions, using the bound on squark mass splittings from the rate of $K^0 \rightarrow \bar{K}^0$ conversion.

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